

Chapter 103 — Physiology and Pharmacology of the Bladder and Urethra

INTRODUCTION

- **Two defining functions** of the lower urinary tract (LUT: bladder + urethra): **urinary storage (continence) and emptying (micturition)**; a third, **immunosurveillance** against ascending pathogens, depends on the urothelium, the immune system and the **urinary microbiome**.
- **Both functions are integrated** across the central nervous system (CNS), peripheral nerves, detrusor smooth muscle (DSM), stroma, suburothelial and intradetrusor interstitial cells, urothelium, urethral smooth muscle, pelvic floor and the **external urethral sphincter (EUS)**.
- Physiology underpins management of **lower urinary tract dysfunction (LUTD)**: urinary incontinence (UI), lower urinary tract symptoms (LUTS), overactive bladder (OAB), bladder pain syndrome/interstitial cystitis (BPS/IC), neurogenic bladder and underactive bladder (UAB).
- **Most data are animal-derived**; limits to translation: **rodent circadian rhythm is opposite** to that of humans; territorial marking by voiding; **animals cannot report urgency or bother**, so surrogates (frequency, urodynamics) are used; **anaesthesia alters reflexes**. Translational (bench ↔ bedside) work remains essential.
- **Classic model**: brain and spinal circuits act as **switches** alternating storage and elimination. But LUTD may arise **outside nerve and muscle**: endocrine (metabolic syndrome, obesity), inflammation, fibrosis, **pelvic ischaemia**.
- **Bladder and urethra must act oppositely**, so single-organ therapy may fail (e.g. boosting contractility in detrusor underactivity fails unless the urethra also relaxes). **Sensory modulation serves both**: reduce afferent signalling for overactivity, augment it for underactivity.
- Management is **symptom-based**; biomarkers, targeted therapy and prevention are the goal.

Concept — The bladder and urethra are a coupled, reciprocal pair: *storage and emptying each need one to do the opposite of the other, which is why afferent (sensory) control, acting on both at once, is the most promising therapeutic lever.*

LOWER URINARY TRACT ANATOMY

- **Bladder body** lies above the ureteric orifices; **base = trigone + bladder neck**. The two regions differ from each other but each is **homogeneous in neuromorphology and neuropharmacology**.
- **Base is laminar**: superficial longitudinal layer beneath the trigone; deeper layer continuous with detrusor, with **small, predominantly circular bundles**.
- **Male bladder neck: complete, competent smooth muscle ring**, with **dense sympathetic noradrenergic supply** that closes it during ejaculation → **antegrade ejaculation**. **Female: no such collar** and sparser adrenergic innervation.
- **Bladder outlet = urethra + pelvic floor**. Urethra contains **smooth muscle and striated muscle (rhabdosphincter, EUS)**; periurethral striated muscle belongs to the pelvic floor complex.
- **Fig. 103.1**: outlet anatomy per **Gosling and Dixon vs Elbadawi** — body/base detrusor, superficial and deep trigone, bladder neck, urethral smooth muscle (lissosphincter), **intramural rhabdosphincter** vs **periurethral striated muscle**.

BLADDER COMPARTMENTS

Urothelium

- **Multilayered: basal, intermediate, apical; about 7 layers;** the apical **umbrella cells** are the largest, sometimes multinucleated, **hexagonal with microvilli** on scanning electron microscopy (SEM) (Figs 103.2B, 103.3).
- **Uroplakins** (unique to umbrella cells) assemble into **plaques** — outer leaflet thicker than inner, hence **asymmetrical unit membrane (AUM)**; the symmetrical areas between plaques are **hinge regions**.
- AUM contains **hexagonal, chainmail-like uroplakin particles** (Fig. 103.4) → **barrier function**. **Escherichia coli type 1 fimbriae** bind uroplakins, initiating the host–pathogen cascade.

Lamina Propria and Vasculature

- **Lamina propria = proposed "functional centre"** coordinating urothelium and DSM: diffuse plexus of **unmyelinated nerves** (contacting urothelium, vessels, DSM), **TRPV1 and P2X3 receptors** on fibres, **interstitial cells (myofibroblasts)** and microvasculature.
- **Corrosion casts** (resin perfusion → digest all other tissue → SEM of vascular tree) show a **mucosal plexus** in the lamina propria and a **subepithelial capillary plexus** directly under the basal urothelium — **the vessels seen at cystoscopy** (Fig. 103.5; adventitial plexus linked by perpendicular vessels traversing the muscularis). Subepithelial flow may be **partly urothelium-regulated**.
- **Vessels must lengthen with filling without raising resistance. Distension reduces blood flow; intramural tension is the principal determinant.** Flow adapts during normal filling until pressure rises; in **low compliance**, pressure rises more and flow falls more.
- **Ischaemia rapidly reduces detrusor contractility. Ischaemia–reperfusion** may damage intramural neurons → **patchy denervation** and altered DSM seen in **detrusor overactivity (DO)**. Animal pelvic ischaemia (iliac intimal balloon injury + high-cholesterol diet) → **DO and structural change**.

Stroma

- **Collagen and elastin in a proteoglycan matrix; fibroblasts** are the main cells. **Passive mechanics = viscoelasticity of stroma + relaxed detrusor.**
- No longer seen as inert: **stroma–muscle interaction likely drives hypertrophy** (in arteries, elastin disruption triggers smooth muscle proliferation; not yet shown in bladder).

Bladder Wall Collagen

- Mostly **outside the muscle bundles; types I, III and IV** predominate.
- **Poorly compliant bladders** (augmentation cohort): ↑ **connective tissue : smooth muscle ratio** and ↑ **type III : type I collagen, even with normal capacity** → poor storage is secondary to connective tissue change, especially ↑ **type III collagen**.
- **Obstruction or denervation (rat)** → DSM hypertrophy → ↓ **collagen concentration** (dilution). **Ageing** → ↓ **smooth muscle relative to collagen** (both sexes), perhaps via ↓ packing density of submucosal collagen.
- In noncompliant bladders, **collagen fibres lose the ability to change tortuosity** → ↓ capacity.

Bladder Wall Elastin and Matrix

- Elastic fibre = **amorphous elastin (about 90%) + peripheral glycoprotein microfibrils; sparse vs collagen** but in all layers.
- **Spinal cord injury (SCI) rat: elastin : collagen ↑ over the first 6 weeks** (compliance ↑, overdilation) → then ↓ **as compliance falls with DO emergence by about 10 weeks** — elastin modulates compliance.

- **Spina bifida:** compliance associates with **urinary MMP-2 and TIMP-2** (matrix metalloproteinase 2 and its tissue inhibitor).
- Non-fibrillar matrix = **proteoglycan-water gel** (glycosaminoglycans, GAGs, on protein cores) — viscous under deformation.

Smooth Muscle

- Bladder body: **fascicles in random directions**, cells coupled into a **functional syncytium** — unlike the discrete circular/longitudinal layers of ureter and gut.
- **Single central nucleus, no striations, spindle-shaped**, several hundred μm long, **5–6 μm diameter**. **Caveolae** (flask-shaped invaginations) associate with sarcoplasmic reticulum (SR); their location and density may set **contractile heterogeneity**.
- **Motor supply: postganglionic parasympathetic fibres** (intramural ganglia exist). **Varicosities** release **acetylcholine (ACh)** and **adenosine triphosphate (ATP)** onto **muscarinic and purinergic** receptors (Fig. 103.6); **not every cell is contacted** — **gap junctions spread excitation**. DSM also has **afferent innervation**.

Concept — Structure sets compliance: *the ratio of contractile protein to collagen (and collagen type and tortuosity) decides whether the bladder stores at low pressure, and ischaemia, obstruction and ageing all act by shifting that ratio.*

OVERVIEW OF THE URETHRA

Male Urethra

- **Four segments: preprostatic (bladder neck) → prostatic → membranous** (through pelvic floor incl. EUS) → **bulbous and penile** (Fig. 103.7, retrograde urethrogram).
- **Male EUS shape: crescent on ventral prostate proximal to verumontanum → horseshoe distal to it → crescent at bulbar urethra.**

Female Urethra

- Runs in the **distal third of the anterior vaginal wall**. Continence aided by tissues beyond the sphincter (Fig. 103.8): **subepithelial vascular plexus → "urethral seal"**. **Female EUS: horseshoe** over the ventral urethra, under **voluntary control**.
- **Continence during raised abdominal pressure:**
 - **Passive pressure transmission** to the proximal urethra.
 - **Guarding reflex:** active EUS contraction. Transmission alone cannot explain the full rise — **urethral pressure rises before cough transmission → active neural mechanism**.
 - **DeLancey "hammock hypothesis":** transmitted pressure compresses the anterior against a **posterior wall kept rigid by pelvic muscle/connective tissue support; pubourethral attachments** and vaginal-fascial connections **compress the urethra against the pubis** and move the bladder neck with voiding.
- **Continence = active muscle tone + passive anatomic coaptation.**

Anatomy Common to Both Sexes

- Fetal dissection: **sphincteric complex develops alike in both sexes**, from detrusor, trigone and urethral muscle of **different embryonic origins**. **Levator ani does not surround the ventral urethra** in either sex — its continence role is questioned.

Fiber Types of Urethral Striated Muscle

SLOW-TWITCH VS FAST-TWITCH STRIATED FIBRES

Feature	Slow-twitch	Fast-twitch
Role	Sustained sphincter tone	Rapid top-up tone on sudden abdominal pressure
Recruitment / fatigue	Slow / slow	Rapid / rapid
Metabolism	Oxidative	Anaerobic
Myosin ATPase	Less	Rich (alkaline-stable)
SR Ca ²⁺ -ATPase	Slow isoform	Fast isoform → rapid relaxation
Where	Levators, urethral sphincter; source of background EMG at urodynamics	Periurethral pelvic floor

- Striated muscle contraction, like smooth, is **Ca²⁺-governed** — via **troponin**.
- **Rhabdosphincter (EUS)** is **intramural**, separate from periurethral pelvic floor muscle; cells **15–20 μm** (smaller than ordinary skeletal muscle).
- **Periurethral pelvic floor** = **mixed fast + slow**; **intramural distal sphincter** = **predominantly slow-twitch**, providing **about 50% of static urethral resistance**.
- **Rhabdosphincter composition**: male **35% fast / 65% slow**; female **13% fast / 87% slow**.
- **Most fast and about a quarter of slow fibres** in the membranous sphincter stain for **nitric oxide synthase (NOS)** in the sarcolemma.
- Pelvic floor exercise or electrostimulation may cure stress urinary incontinence (SUI) by **converting fast- to slow-twitch fibres** (speculative).
- **EUS may contain noradrenergically innervated smooth muscle**: hypogastric stimulation evokes EUS potentials that **persist after α-blockade** → probably striated in origin.

Concept — Tone vs reserve: *slow-twitch oxidative fibres hold resting closure (hence the female sphincter's slow-fibre dominance), fast-twitch fibres supply the brief extra squeeze on coughing — and continence always combines active tone with passive anatomical support.*

UROTHELIAL PHYSIOLOGY

- Primary role: **relatively impermeable barrier**; also **sentinel defence against uropathogens**, **afferent signalling and modulation of DSM contractility**.

Barrier Function

- Permeability depends on **passive diffusion**, **osmotically driven diffusion**, **active transport** and **membrane inertness**.
- **Finite passage has long been known** (Kaupp, 1856: 12-hourly vs hourly voiding altered urine composition). **Passive permeability to most solutes**; in vivo rat urothelium passes **urea, Na⁺, K⁺, Cl⁻**, so the bladder may **modify final urine**, depending on hydration and dietary protein.
- **Water crosses via aquaporins**; human water permeability about **6.5 × 10⁻⁵ cm/s** (tritiated water absorption); **no direct human diffusive permeability measure yet** — one could phenotype LUTD with a leaky urothelium.
- **Loss of umbrella cells (cystitis models)** → **↑ water and urea permeability**; urinary solute leak into the lamina propria likely **causes cystitis symptoms**, worsened by **distension** (intermediate/basal cells are no real barrier). Similar in **infective and radiation cystitis**.

- **Osmotic gradient maintained:** plasma **about 300 mOsm/kg** vs urine **100–1500 mOsm/kg** — unnoticed normally, **symptomatic once inflamed (BPS/IC)**.
- **SCI and myelodysplasia** → chronic bacteriuric cystitis. **Hyperosmolar instillation (rat)** → stronger contractions, BP rise, **neurogenic inflammation (Evans blue extravasation) blocked by capsaicin** → hyperosmolality **excites C-fibre afferents**; enhanced after **dimethyl sulfoxide (DMSO)** pretreatment (simulated cystitis).
- **Tight junction (TJ) proteins:** **zona occludens-1 (ZO-1), occludin, claudin-4, -8, -12** — block **paracellular flow** (Fig. 103.9: TJs, adherens junctions, desmosomes, gap junctions). With stretch, **small-molecule permeability is unchanged but transepithelial resistance (TER) falls tenfold** → **↑ ionic paracellular transport via claudin pathways**.
- **GAG layer — controversial:**
 - GAG is **100% polysaccharide**, made enzymatically (not template-coded) → **heterogeneous**.
 - Theory arose from **acid-injured rabbit bladder**: **↑ bacterial adherence**, normalising within 24 h → "secretory" antiadherence factor, attributed to GAG.
 - Later shown to be a **glycoprotein (mucin MUC-1)**; MUC-1-transfected cells showed **no change in water/urea permeability**. **Chondroitin sulfate** is the main human GAG.
 - **Verdict:** GAG may matter for **bacterial antiadherence and blocking macromolecules**, but is **not proven to be the primary impermeability barrier**.
- **Trigone and detrusor urothelium are alike**; the **proximal urethra changes** to stratified/columnar epithelium without urothelial differentiation markers.
- **Urethral chemosensory "brush" cells (paraneurons):** luminal, with microvilli, like **gut enterochromaffin or tracheal brush cells**; detect noxious stimuli or bacteria by the **taste-transduction pathway**, release **ACh and serotonin** onto nearby sensory nerves → alter bladder function. **Intraurethral serotonin** → **↑ visceromotor response (VMR)** and afferent firing; **sacral afferents respond to urethral flow**.

Urethral Tone

- **Smooth muscle, striated muscle and lamina propria all contribute** to the urethral pressure profile.
- **Nicotinic (striated) blockade** reduces tone **rarely by >40%** → smooth muscle matters; **α-blockade** lowers urethral pressure by **about one third**; **little cholinergic contribution**.
- Despite a **posteriorly open horseshoe**, EUS pressure during filling rises **uniformly like an iris**; **noradrenaline or hypogastric stimulation augments it** → adrenergic role in EUS.
- Urethral stroma: **longitudinal collagen and elastin**. **Vascular filling of the lamina propria aids continence** (magnitude unknown); **oestrogen ↑ urethral blood flow**.
- **Impaired arterial supply ↓ urethral pressure:** early fall = **↓ vascular filling**; later = **hypoxic effect on smooth muscle**.
- **Periurethral collagen in SUI — divergent:** premenopausal SUI: more collagen and thicker fibrils; postmenopausal: **no difference**; others report **↓ collagen** and **↓ collagen I : III**.

Ionic Transport

- **Apical membrane: high resistance; basolateral about tenfold lower.** **Active Na⁺ transport** occurs.
- Apical Na⁺ channels (surface and subapical vesicles) are **amiloride-sensitive and aldosterone-responsive**; amiloride-insensitive cation channels are **degradation products**, cleaved by **urinary serine proteases (kallikrein, urokinase, plasmin)**.
- Basolateral **Na⁺/K⁺ exchanger** extrudes Na⁺ (cell interior negative); **K⁺ and Cl⁻ channels, Na⁺/H⁺ and Cl⁻/HCO₃⁻ exchangers** restore cell volume when serosal osmolality rises.

- **Role of the apical Na⁺ channel unknown:** channel degradation with filling may **signal fullness**, or regulate **apical membrane insertion/removal** during expansion.

Sensor-Transducer Function of the Urothelium

- Urothelial cells behave like **sensory neurons (nociceptors, mechanoreceptors)**. "Sensor molecules" expressed: receptors for **bradykinin**, **neurotrophins (TrkA, p75)**, **purines (P2X, P2Y)**, **noradrenaline (α , β)**, **ACh (nicotinic, muscarinic)**, **protease-activated receptors**; **ENaC** (amiloride-sensitive mechanosensitive Na⁺ channel); **TRPV1, TRPV2, TRPV4, TRPM8** (transient receptor potential channels).
- Activated by mechanical or chemical stimuli, they **release NO, ATP, ACh and substance P (SP)**, acting on nearby afferents **directly or via suburothelial myofibroblasts** (gap-junction linked) → chemical coupling to sensory nerves. (Carbachol causes dose-dependent ↑ intracellular Ca²⁺.)
- **Nitric oxide (NO):** released especially in **inflammation** (by calcium ionophore, noradrenaline, capsaicin, SP); **urothelium removal cuts adrenergic NO release by about 85%**; cultured urothelium of several species (incl. human) expresses **TRPV1**; afferent NOS ↑ in chronic inflammation. NO barely affects detrusor but **inhibits afferent Ca²⁺ channels** → **net inhibition of afferent activity** (NO scavenging or NOS block ↑ bladder activity).
- **ATP from stretched urothelium** → **suburothelial P2X3 afferents** → fullness and pain; **P2X3-null mice show hyporeflexia**. Analogy: **airway epithelium modulating submucosal nerves**. ATP also acts **autocrine (P2X, P2Y)** to trigger **exocytosis/membrane trafficking**. Human urothelium releases ATP to muscarinic/purinergic stimulation, causing **mucosal contraction**; ageing ↑ ATP release.
- **Prostaglandins (PGs):** urothelium is the **main PG source**; roles = **detrusor regulation and cytoprotection** (PGs treat **haemorrhagic cystitis**). Human urothelium: **6-keto-PGF1 α > PGE2 > PGF2 α > thromboxane B2**, plus **prostacyclin (PGI2)**; production ↑ **greatly in inflammation**. Ureteric PGs may regulate **peristalsis** and **limit luminal clot**.
- **Muscarinic receptors on urothelium at high density**; basal ACh release ↑ by stretch and ageing → modulates afferents and DSM.
- **Urothelium-derived inhibitory factor:** ↓ muscarinic contraction; **identity unknown** — not NO, PG, prostacyclin, adenosine, catecholamine, γ -aminobutyric acid (GABA) or apamin-sensitive SK-channel action; attenuated in fetal bladder outlet obstruction (BOO).

Suburothelial Interstitial Cells

- **Myofibroblasts beneath the basal urothelium:** **vimentin+** and **α -smooth muscle actin+**, **desmin-**; linked by **connexin 43 (Cx43)** gap junctions; **close to C-fibre endings** → a functional network (Fig. 103.10: stretch-released ATP/ACh act via **P2Y, M2/M3, TRPV1** on interstitial cells → afferents).
- **ATP** → inward current + ↑ Ca²⁺; express **P2Y6** and **M3**. In human **idiopathic DO (IDO)**, **M2/M3 expression on these cells correlates with urgency score**.
- **NO donor sodium nitroprusside (SNP)** damps their ATP responses → **cyclic guanosine monophosphate (cGMP)-dependent inhibition**.

KEY POINTS: UROTHELIUM

- **Uroplakins and TJ proteins** underpin barrier function.
- **Uroplakins are the attachment site of type 1-piliated uropathogenic E. coli.**
- **GAG layer** may aid antiadherence; **no definite evidence of impermeability function.**
- **Urothelial cells release and respond to neurotransmitters.**

- **Myofibroblasts mediate urothelium–afferent interaction.**

Concept — The urothelium is a sensory organ as well as a wall: *it detects stretch and chemistry and signals to afferents (ATP excites, NO inhibits, myofibroblasts relay), so a damaged barrier produces symptoms twice over — by solute leak and by altered signalling.*

SMOOTH MUSCLE PHYSIOLOGY

- DSM must contract for efficient emptying; **understanding its quiescence informs OAB, neurogenic bladder and underactive detrusor.**
- **Differs from striated muscle:** autonomic (not somatic) motor supply, **modulation by hormones, paracrine NO and urothelial factors**, and **spontaneous rhythmic activity without neural input**; efficient voiding still needs neural control.

TABLE 103.1 COMPARISON OF THE PROPERTIES OF SKELETAL AND SMOOTH MUSCLE

Property	Skeletal muscle	Smooth muscle
Cells	Long cylinders, many nuclei	Spindle, single nucleus
Maximum size (length × diameter)	30 cm × 100 µm	200 µm × 5 µm (text: several hundred µm × 5–6 µm)
Striations	Yes	No
Ultrastructure	Sarcomeres ; no intermediate filaments	No sarcomeres; intermediate filaments, dense bodies
Motor innervation	Somatic	Autonomic
Contracture	Phasic	Mostly tonic , some phasic
Contractile activation	Disinhibition of tropomyosin ; sliding filaments; rapid	Active myosin phosphorylation ; sliding filaments (?); "latch state"
Ca ²⁺ regulation	Rapid influx via T tubules	Voltage- and receptor-operated channels; internal stores
Basal tone	Neural activity	Intrinsic and extrinsic factors
Force regulated by hormone	No	Yes

After Chacko et al., 1999.

- **What it tests:** skeletal muscle is **switched on by removing an inhibitor (troponin–tropomyosin)**; smooth muscle is **switched on by phosphorylating myosin** — so it is slower, sustains tone economically (latch) and is open to hormonal and local modulation.
- **Trap:** the lengths differ between table (200 µm) and text ("several hundred µm") — both describe cells of a few hundred micrometres.

Contractile Proteins

SMOOTH MUSCLE FILAMENTS

Filament	Diameter	Composition	Role
Thick	15 nm	Myosin II	Cross-bridges
Thin	6–8 nm	Actin + tropomyosin (TM) + caldesmon (CaD)	Regulated track

Filament	Diameter	Composition	Role
Intermediate	10 nm	Desmin, vimentin	Modulate response; anchor at dense bodies

- **Myofibrils run obliquely in a lattice** (no sarcomeres); thin and intermediate filaments anchor to **dense bodies** in cytoplasm and **dense bands** between caveolae; junctional complexes link neighbours → **syncytial contraction** (Fig. 103.11).
- **Myosin II = two intertwined myosin heavy chains (MHCs)**: rod/coil tail, hinge, globular heads (Fig. 103.12). **Splicing**: 5' end → **SM-A / SM-B**; 3' end → **SM-1 / SM-2**. **Bladder is almost 100% SM-B — higher ATPase, faster actin movement** than SM-A.
- Each head carries **two myosin light chains**: **MLC20** (regulatory — **its phosphorylation drives cycling** via myosin ATPase) and **MLC17** (two splice variants of one gene).
- **Actin is dynamic**: **actin polymerisation at the cell periphery** is a seminal event, recruiting proteins linking actin to **integrins at adhesion junctions**, stiffening the membrane for force transmission — **regulated separately from cross-bridge cycling** (Fig. 103.13).
- **CaD and TM gate myosin access to actin** (Fig. 103.14). **CaD** (isolated 1981 from chicken gizzard): **h-CaD in all smooth muscle, I-CaD in non-muscle cells**; **inhibits actomyosin ATPase and motility**, enhanced by TM; modulated by **calmodulin (CaM)** — the protein that binds Ca^{2+} to start contraction.

Actinomyosin Cross-Bridge Cycling

- Sequence (Box 103.1): **action potential (AP) → $\uparrow [\text{Ca}^{2+}]_i$ → Ca^{2+} binds CaM → CaM activates myosin light chain kinase (MLCK) → MLCK phosphorylates MLC20 (ATP → ADP) → phospho-myosin binds actin → force; myosin light chain phosphatase (MLCP) dephosphorylates MLC20 → detachment → relaxation**, as Ca^{2+} falls.
- **Tone = Ca^{2+} balance + MLCK : MLCP balance**. Inhibiting MLCP raises force without any rise in Ca^{2+} = **calcium sensitisation**.
- **RhoA** (small GTPase): **inactive with GDP, active with GTP**. Regulators: **GEFs** (GDP → GTP, activate), **GAPs** (GTP → GDP, inactivate), **GDIs** (bind RhoA-GDP, block GEFs and membrane translocation).
- **RhoA-GTP activates Rho kinase (ROK)** at the membrane → **ROK phosphorylates and inactivates MLCP**; ROK (and protein kinase C, PKC) also phosphorylate **CPI-17**, which then **binds and inhibits MLCP** → contraction (Fig. 103.15).

BOX 103.1 DETRUSOR SMOOTH MUSCLE CONTRACTION — SEQUENCE

1. **Ca^{2+} binds calmodulin**, activating it.
2. **CaM activates MLCK**.
3. **MLCK transfers phosphate from ATP to myosin**, allowing interaction with actin.
4. **Relaxation follows the fall in intracellular Ca^{2+}** .

Membrane Electrical Properties and Action Potentials

- DSM is **excitable**; contractility depends on membrane potential. **Single-cell patch data ignore the syncytium and interstitial pacemakers**.

ION CONCENTRATIONS (MEQ/L)

Ion	Intracellular	Extracellular
Na^+	5–15 (low)	130–140 (high)
K^+	130–140 (high)	3–5 (low)

Ion	Intracellular	Extracellular
Cl ⁻	4–30 (low)	105–115 (high)

- Membrane potential follows the **Goldman–Hodgkin–Katz equation** (weighting each ion's gradient by its permeability). **At rest the membrane is K⁺-permeable only**, so resting potential approaches the **K⁺ reversal potential (about -90 mV)**; reversal potentials: **Na⁺ about +65 mV, Cl⁻ about -85 mV**.
- **Human detrusor resting potential -50 to -60 mV** (cultured cells more depolarised, about -28 mV).
- **Detrusor AP (Fig. 103.16): phase 0 slow depolarisation → phase 1 fast upstroke (Ca²⁺ inward current, I_{Ca}) → phase 2 repolarisation → phase 3 hyperpolarisation (both K⁺ outward current, I_K); continuous current gives AP trains.**
- **IK channels: SK and BK** (small- and large-conductance Ca²⁺-activated K⁺) and **Kv** (voltage-gated). **Blocking any → ↑ contractility and spontaneous activity. BK is the most critical:** sets membrane potential → governs **L-type voltage-dependent Ca²⁺ channel (VDCC)** opening → [Ca²⁺]_i → contraction.

Excitation-Contraction Coupling

- **ACh (postganglionic parasympathetic) → M3 → ↑ [Ca²⁺]_i by (1) extracellular entry through L-type VDCCs and (2) SR release via Gq → inositol trisphosphate (IP3).**
- **Gq cascade (Fig. 103.17): M3 → Gαq → phospholipase C (PLC) → PIP2 cleaved → IP3 (cytosolic; SR IP3 receptor → Ca²⁺ release) + diacylglycerol (DAG, membrane; activates PKC).** Second-messenger families: **IP3/PKC** and **cAMP/PKA** (cyclic adenosine monophosphate/protein kinase A).

G-PROTEIN COUPLING OF RELEVANT RECEPTORS

Coupling	Receptors	Downstream
Gq	Odd muscarinics M1, M3, M5; α1-adrenoceptors; angiotensin II; histamine H1; 5-HT2	PLC → IP3 + DAG → Ca ²⁺ release, PKC
Gi/o	Even muscarinics M2, M4	Inhibit adenylyl cyclase → ↓ cAMP
Gs	β-adrenoceptors	↑ cAMP → PKA → relaxation

- G-protein-coupled receptor work won the **2012 Nobel Prize in Chemistry (Kobilka, Lefkowitz)**. **M3 knockouts prove M3 mediates contraction** (mouse, human).
- **Dogma challenged:** in mouse DSM after electric field stimulation (EFS; purinergic component blocked with α,β-methylene ATP), **contraction depended only on L-type VDCC Ca²⁺ entry** (blocked by atropine and diltiazem, **not** by IP3 or PLC inhibition).
- **Purinergic excitation in disease: obstructed and OAB human bladders show atropine-resistant contractions blocked by α,β-methylene ATP; normal human bladders have none. Purinergic signalling ↑ in the ageing bladder.**
- **P2X1 + M3 both contract mouse DSM, but purinergic activation suppresses muscarinic coupling → explains detrusor hyperactivity with impaired contractility (DHIC; Resnick and Yalla, 1987):** purinergic ↑ afferent drive (urgency/overactivity) + ↓ muscarinic contraction (incomplete emptying) → **blocking purinergic signalling could treat DHIC.**

Calcium Signaling in Detrusor Myocyte

VOLTAGE-DEPENDENT CA²⁺ CHANNELS IN DSM

Feature	L-type (long-lasting)	T-type (transient)
Activation	AP depolarisation	Smaller depolarisations
Conductance	Larger	Smaller

Feature	L-type (long-lasting)	T-type (transient)
Blockers	Nifedipine (dihydropyridine), diltiazem (non-dihydropyridine), verapamil (phenylalkylamine)	No specific blocker yet
Role	Main contraction trigger	Finer control; possibly pathology

- Gene family **CaV/CACNA1**. Ca^{2+} imaging shows a "**calcium flash**" then shortening.
- **Calcium-induced calcium release (CICR)**: VDCC entry → SR release via **ryanodine receptors (RyRs)** → "**calcium sparks**" (first seen in arterial muscle) → open nearby **BK (KCNMA1)** → K^+ efflux → **hyperpolarisation, relaxation**; sparks cause **spontaneous transient outward currents (STOCs)**. Sparks are **not tightly coupled to VDCCs in detrusor (SR farther from membrane) unlike cardiac muscle** — resembles arterial muscle.
- **Ca^{2+} must return to submicromolar levels** by **P-type ATPases** (autophosphorylated aspartate) and other routes (Fig. 103.18): (1) $\text{Na}^+/\text{Ca}^{2+}$ exchange, (2) plasma membrane Ca^{2+} -ATPase, (3) **SERCA into SR**, (4) Ca^{2+} -binding proteins, (5) **mitochondrial uptake**.
- **Store-operated Ca^{2+} entry (SOCE; capacitive entry)**: SR depletion sensed by **stromal interaction molecule (STIM)** → STIM clusters tether **ORAI tetramers** → **calcium release-activated channel (CRAC; current ICRAc)**. Not yet measured in detrusor, but likely.

Propagation of Electrical Responses

- **Cx43 gap junction**: monomer with **four transmembrane domains** (both termini intracellular) → **six form a hemichannel** with a central pore → **two hemichannels dock** → ionic coupling (Fig. 103.19); shown in cultured human myocytes by **double whole-cell patch** and Western blot. **Cx43 and Cx45** expressed in human detrusor.
- **Yet detrusor is poorly coupled**: **no fused tetanic contractions**, high tissue impedance — probably **protective, preventing synchronous activation during filling**; coupling within bundles does exist (measurable length constant).
- **Coupling falls postnatally**: neonatal **large, low-frequency coordinated** activity → adult **small, high-frequency irregular** activity. **Altered coupling may generate uninhibited contractions** in overactive and ageing bladders.

Detrusor Interstitial Cells

- **Normal bladder may be spontaneously active**; exaggerated activity may cause DO (in partial outlet obstruction, local contractions are **more frequent and coordinated**). Source unresolved: **myocytes themselves or interstitial cells as pacemakers**.

INTERSTITIAL CELL POPULATIONS

Population	Markers / receptors	Proposed role	Pharmacology
Suburothelial myofibroblasts	Vimentin, α -SMA, Cx43; P2Y6, M2/M3, TRPV1	Relay urothelium → afferents	NO/SNP inhibit via cGMP
Intradetrusor c-KIT+ cells	c-KIT, M3; COX → PGE2	Pacemaker or modulator of Ca^{2+} transients (not contractile)	Imatinib ↓ spontaneous contractions (guinea pig; human OAB strips with ↑ c-KIT cells); PDE5 inhibitors → ↑ cGMP → ↓ bundle synchrony ; EP antagonists ↓ activity

Population	Markers / receptors	Proposed role	Pharmacology
PDGFR α + cells	Platelet-derived growth factor receptor- α ; P2Y1	Inhibitory: purine-induced prolonged relaxation ; a fibroblast lineage	—

- Interstitial cells lie **along bundle boundaries** (Fig. 103.20), are **spontaneously active**, fire **Ca²⁺ waves** via **M3**; but spontaneous transients arise **independently in muscle and interstitial cells**, so they may **modulate transmission** rather than pace.
- "Dual receptive field" for purines: **excitatory P2X1 on smooth muscle** vs **inhibitory P2Y1 on PDGFR α cells** → fine control.

KEY POINTS: SMOOTH MUSCLE MECHANICS

- **Muscarinic receptors contract detrusor** in response to parasympathetic ACh, via **Ca²⁺ entry through Ca²⁺ channels**.
- Ca²⁺ triggers all muscle, but in smooth muscle the **activation mechanism differs**; the response is **slower and longer lasting** than skeletal or cardiac.
- The **normal bladder may be spontaneously active**; exaggerated spontaneous contraction may contribute to OAB; **interstitial cells** are proposed pacemakers.

Concept — Contraction is a phosphorylation balance: *Ca²⁺ entry (mainly L-type) switches MLCK on, MLCP and BK channels switch the cell back off, and nearly every pathology in this section — DO, DHIC, calcium sensitisation — is a shift in that kinase-phosphatase or excitation-brake balance.*

BLADDER MECHANICS

Urinary Storage (Filling)

- **Wall composition:** besides muscle, **about 50% collagen, 2% elastin**; collagen $\uparrow$ after injury, obstruction or denervation. **Contractile protein > collagen** → **compliant**; $\uparrow$ collagen → $\downarrow$ compliance.
- **Filling geometry:** lamina propria thins faster than muscle; bundles shift from **top-to-bottom** to **side-to-side**; **coiled type III collagen straightens**, becoming taut and **parallel to the lumen** → storage at low pressure.
- **Compliance C = $\Delta V / \Delta P$.** **Slow physiological filling keeps pressure <10 cmH₂O**; compliance is **fill-rate dependent** (pressure drops when urodynamic filling slows/stops). In women with SUI (50 mL/min), detrusor pressure rose **only a few cmH₂O** to capacity.
- **Pressure-volume curve depends on** (1) passive collagen, elastin and muscle, (2) active muscle, (3) geometry. **Low compliance (steep curve) may reflect:** **fast filling**; **more collagen/less elastin**; **smooth muscle hyperactivity**; or combinations.
- **Acute spinal transection does not alter compliance**; **pelvic nerve transection reduces it**. Compliance involves an **active afferent-driven CNS loop inhibiting spontaneous contractility** during filling.
- **Broad length-tension relationship**; viscoelasticity shapes wall tension. **Spontaneous activity** common in small-mammal strips and seen in human, but **fused tetanic contractions (as in gut or uterus) almost never occur normally**.
- "Dynamic elasticity": compliance is **continuously modulated** during filling; **ex vivo pulsatile compression $\uparrow$ capacity** (pig) — a possible way to reduce urgency from micromotions.

Voiding Mechanics

- $P_{det} = P_{ves} - P_{abd}$ (detrusor = intravesical – abdominal). Micturition = **neurally driven detrusor contraction raising P_{det} without P_{abd} change.**
- **P_{det} alone does not measure contraction strength:** muscle spends energy on **force** ($\rightarrow P_{det}$) or **shortening velocity** ($\rightarrow$ flow, Q); **trade-off $\approx$ constant power $W = P_{det} \times Q$.**
- **During voiding P_{det} reflects outlet resistance.** With a wide-open urethra, little pressure is needed: **stress-incontinent women void at P_{det} at Q_{max} about 19 cmH₂O; healthy men about 60–70 cmH₂O.**
- **Low voiding pressure in a woman $\neq$ impaired contractility; male obstruction nomograms must not be applied to women without validation.**

Motor Sensory Network in Detrusor Muscle

- **Sensory fibres lie beside myocytes and intradetrusor interstitial cells** (Fig. 103.21, with intramural and external ganglia); **spontaneous micromotions trigger afferent signals** $\rightarrow$ $\uparrow$ activity may be **perceived as urgency.**
- Explains how **antimuscarinics (tolterodine) and the β_3 agonist mirabegron** $\downarrow$ non-voiding contractions (partial obstruction model) and so $\downarrow$ sensory input (afferent output not measured). **Micromotions seen in women with sensory urgency.**
- **Efferent and afferent functions are inseparable.**

Concept — Pressure is not strength: *detrusor power is shared between pressure and flow, so P_{det} during voiding reports outlet resistance, and during filling low pressure depends on wall structure plus an active afferent–CNS brake on spontaneous contraction.*

NEURAL CONTROL OF THE LOWER URINARY TRACT

Peripheral Nervous System

THE THREE PERIPHERAL NERVE SUPPLIES (FIG. 103.22)

Feature	Parasympathetic	Sympathetic	Somatic
Origin	Sacral (S2–S4) intermediate grey: sacral parasympathetic nucleus	Rostral lumbar (thoracolumbar) cord	Sacral ventral horn: Onuf's nucleus
Nerve	Pelvic	Hypogastric	Pudendal
Bladder body	Excites (ACh $\rightarrow$ M3; ATP)	Inhibits (noradrenaline $\rightarrow$ β_3)	—
Bladder base/urethra	Relaxes urethra (NO)	Contracts (α_1)	Contracts EUS (ACh $\rightarrow$ nicotinic)
Net role	Voiding	Storage	Storage (voluntary closure)

- All three nerves are **mixed afferent + efferent.**
- **Parasympathetic:** preganglionic axons via **ventral roots** release ACh in **pelvic plexus and intramural ganglia**. Postganglionic neurons lie in the **detrusor wall as well as the pelvic plexus** $\rightarrow$ **cauda equina or pelvic plexus injury decentralises but may not fully denervate**; intramural afferent–efferent interconnection is possible.
- **Sympathetic route:** lumbar cord $\rightarrow$ **sympathetic chain ganglia** $\rightarrow$ **inferior splanchnic nerves** $\rightarrow$ **inferior mesenteric ganglia** $\rightarrow$ **hypogastric nerves** $\rightarrow$ **pelvic ganglia.**
- **Somatic:** EUS motoneurons in **Onuf's nucleus** (lateral border of ventral horn; Fig. 103.23), with **transverse dendritic bundles** projecting to the lateral funiculus, intermediate grey and toward the central canal.

Afferent Pathways

Overview: properties of afferent neurons

- **Most time is spent storing:** reflexes suppress DSM and contract the sphincter; micturition reverses this. **Sensory signals drive the switch** and generate **fullness and pain**. Dysfunction → OAB, BPS/IC (urgency, pain, UI).
- Current drugs mainly **reduce contraction**, yet **symptoms reflect dysregulated storage sensation** rather than exaggerated contraction → **targeting afferents is rational**.

Pathways to the spinal cord

- Afferents travel in **pelvic, hypogastric (lumbar splanchnic) and pudendal nerves**. Cell bodies: **pelvic and pudendal → sacral dorsal root ganglia (DRG)**; hypogastric → **rostral lumbar DRG**. Second-order spinal neurons mediate **spinal reflexes and ascending pathways**.
- **Pelvic afferents** (volume and contraction) = **myelinated Aδ + unmyelinated C**. In **SCI (and probably inflammation) C fibres are recruited** into a new pathway → **urgency urinary incontinence (UI)** and possibly pain (Figs 103.24, 103.25).
- **Sensation:** awareness and desire to void map to bladder distension and contraction; **sensation of imminent micturition may arise from the urethra**. Filling produces **episodic sensory bursts of rising intensity**, the last perceived as **urgency**. Afferent firing during contraction **reinforces the central drive**.
- **Receptor types:** "**in-series tension receptors**" (respond to distension and contraction; cat, rat) vs **volume receptors** (distension only, possibly urothelial stretch; rat); many rat **C afferents are stretch receptors unresponsive to contraction**.

Functional properties of bladder afferents

- **Human: only unmyelinated nerves in urothelium/suburothelium; small myelinated fibres with muscle.** Lamina propria plexus densest at bladder neck and proximal urethra, absent at the dome; muscle innervation uniform.
- **Most afferents are mechanosensitive** with thresholds from physiological to noxious; **low threshold = small myelinated; high threshold = generally unmyelinated**. Others are **chemoreceptors** (hypertonic saline, capsaicin, ATP) or "**silent**" afferents sensitised by inflammation → pain.

TABLE 103.2 BLADDER AFFERENT PROPERTIES

Fibre type	Location	Normal function	Effect of inflammation
Aδ (finely myelinated)	Smooth muscle	Sense fullness (wall tension)	Discharge at lower pressure threshold
C (unmyelinated)	Mucosa	Respond to stretch (volume sensors)	Discharge at lower threshold
C (unmyelinated)	Mucosa, muscle	Nociception to overdistension; silent afferents	Sensitive to irritants; become mechanosensitive → new afferent pathway

- **Mouse pelvic nerve — four classes: serosal, muscular, muscular/urothelial, urothelial; low-threshold muscular afferents are the majority (65–80%),** plus high-threshold; sensitised by inflammatory soup (bradykinin, 5-HT, PG, histamine at pH 6.0).
- **Guinea pig:** low-threshold "**antenna-like**" muscle endings = **tension receptors** (respond to stretch and contraction); **high-threshold endings in deep muscle/serosa near blood vessels**, sensitised by inflammation.
- **Species differences:** Aδ/C conduction cut-off 2 m/s (cat) vs 1.3 m/s (rat). Cat: Aδ = **low-threshold mechanoreceptors**; C = **mechanoinsensitive "silent C fibres"** sensitised by **high K⁺, low pH, high osmolality, capsaicin** (Fig. 103.26), after which silent afferents become mechanoreceptive.

- **Bladder neck and proximal urethra have the densest innervation**; urethral afferents lie between muscle fibres, around vessels, in the urothelium and a dense suburothelial plexus. **Striated sphincter: sparse afferents; muscle spindles (group Ia) absent or scarce.**
- **Urethral paraneurons** (like tracheal brush cells) may release **neuropeptides or serotonin** onto sensory nerves → urethral reflexes. **Flow-responsive sacral afferents behave like cutaneous afferents** — relevant to **restoring emptying after SCI**.
- **Fig. 103.26 — sensitisation**: urothelial ATP → P2X; axoplasmically transported neuropeptides released on distension; **neurokinin A → NK2 autoreceptors**; **high urinary K⁺, high osmolality, low pH**, inflammatory mediators; muscle force; **nerve growth factor (NGF) → TrkA**.

Modulators of afferent sensitivity

- **Stimulus–response function is shaped by the ending's environment**: contraction distorts endings; connective tissue transmits or dissipates energy (**rapid vs slow adaptation**); urothelial and lamina propria mediators alter firing.
- **Mediators** (neurotrophins, purines, proteases, prostanoids, cytokines — from mast cells, fibroblasts, neurons) act by **three processes**: (1) **directly open ion channels**; (2) **sensitise without stimulating (hyperexcitability)**; (3) **change phenotype long-term** (neurotrophins alter channel/receptor/mediator expression).
- **Sensitisation = leftward shift of stimulus–response curve** (greater barrage per stimulus). **Peripheral sensitisation is rapid and short-lived**; with maintained injury, **gene expression changes → central sensitisation**.
- **Reduced afferent sensitivity → impaired emptying: diabetic cystopathy** (↓ sensation, ↑ residual; ischaemia, glucose metabolism, free radicals); **aged rats ↓ volume sensitivity and neuropeptides**; **aged humans ↓ brain response to filling**.

Nitric oxide

- **Major inhibitory transmitter relaxing urethral smooth muscle during micturition**; also **inhibits bladder afferents**.
- NOS inhibitors **do not alter normal micturition**, but **spinal NOS inhibition ameliorates irritation-induced DO**. **L-NAME ↑ afferent filling response by about 50%**, reversed by **L-arginine**; L-arginine also **reverses acrolein (cyclophosphamide metabolite) hypersensitivity (BPS/IC model)**.
- **Signal via cGMP** (Fig. 103.27): **soluble guanylyl cyclase ← NO**; **particulate guanylyl cyclase ← C-type natriuretic peptide**; membrane adenylyl cyclases ← **Gsα and forskolin**; soluble adenylyl cyclase ← **HCO₃⁻ and Ca²⁺**. **Phosphodiesterases (PDEs) hydrolyse cAMP/cGMP**; **PDE5 terminates NO action → PDE5 inhibition ↓ afferent activity after SCI** → target for **BPS/IC and OAB**. Injury/inflammation ↑ inducible NOS.

Purinergic signalling

- **Stretch → urothelial ATP → myofibroblasts and afferents** (Fig. 103.28; ATP ↑ with stress, injury, inflammation, infection). **P2X2 and P2X3 on unmyelinated afferents**; mechanotransduction via **P2X2 and P2X2/3**.
- **Normal thresholds reflect an ATP (excitatory) : NO (inhibitory) balance**. ATP ↑ in DO and BPS/IC. **Diabetic DO model: ↑ ATP, normal NO**; **sugar-induced UAB model: ↑ NO, normal ATP**.

*Transient receptor potential cation channels***TRP CHANNELS IN THE BLADDER**

Channel	Activators	Location	Role / therapy
TRPV1	Heat >43 °C, protons, capsaicin	C-fibre nociceptors; suburothelial and muscle plexuses; urothelium	Inflammatory hypersensitivity; desensitised by resiniferatoxin (RTX) and capsaicin
TRPV4	Stretch/pressure, hypotonicity, warmth >27 °C, phorbol esters	Urothelium (adherens junctions) → ATP release; C fibres	Knockout → impaired voiding; inhibition improves cystitis
TRPA1 (only ankyrin TRP)	Mustard oil (allyl isothiocyanate), allicin, cinnamaldehyde, acrolein; bradykinin via G proteins	C fibres with CGRP; urothelium	Agonists → overactivity; ↑ in SCI; blockade normalises reflexes
TRPM8	Cold <23 °C, menthol, icilin	Afferents; DRG with CGRP, IB4	Bladder cooling reflex / ice-water test; blocker ↓ frequency

- **RTX:** principal active ingredient of **euphorbium** (latex of *Euphorbia resinifera*); **isolated 1975**; recognised **1989** as an **ultrapotent capsaicin analogue** that **desensitises without prior excitation**. Intravesical RTX in SCI DO → nerve staining changes and cystometric improvement. Capsaicin/RTX helped IC pain in small studies, but a randomised trial of intravesical RTX in IC was negative.
- **Bladder cooling reflex:** cold saline → contraction below normal threshold; **once taken to indicate a supraspinal lesion**; used for OAB and BOO.

Cannabinoids

- **Cannabinoids in Multiple Sclerosis (CAMS) study:** cannabis extracts improved UI and DO in multiple sclerosis (MS).
- **CB1 > CB2** in human urothelium and detrusor; **CB1+ urothelial fibres ↑ in BPS/IC and IDO**; **CB1 agonist ↓ distension firing, especially high-threshold (noxious) afferents**. Others: **CB2 predominates** in urothelium, suburothelium and sensory nerves; **CB2 agonists ↓ nerve-evoked contractions**. **Net: modulatory role in afferent signalling.**

Opioid peptides

- **μ-opioid agonism ↓ bladder sensation and ↑ sphincter tone → retention risk** (postoperative). Tramadol ↑ capacity and compliance (animals).
- **Short half-life opioid (fentanyl) → less retention than morphine**. COX-2 inhibitor co-therapy ↓ opioid use, nausea and constipation but not retention.

Pelvic organ interactions: crosstalk between bladder and bowel

- **Irritable bowel syndrome (IBS) ↔ LUTS; BPS/IC ↔ bowel symptoms → cross-organ sensitisation → chronic pelvic pain syndrome**. **Colitis ↑ bladder contraction frequency; cystitis sensitises bowel**; also among uterus, urethra, vagina and prostate.
- **Mechanisms (Fig. 103.29): (1) dichotomising afferents** in DRG (axon reflex, TRPV1/P2X3+); **(2) spinal convergence** (dorsal root reflexes; laminae VI–VII output neurons); **(3) brain convergence**. **Viscerosomatic convergence explains referred pain** (angina); convergence explains **poorly localised pelvic pain**.

KEY POINTS: AFFERENT PATHWAYS

- Rat pelvic nerve contains **in-series tension receptors, volume receptors and silent afferents (incl. nociceptors)**.
- **Intravesical irritants lower pressure thresholds** of most endings, including high-threshold ones.

- **Much of the C-fibre population is silent** to normal distension but **becomes mechanosensitive** after chemical mediators.
- In inflammation and other pathology, **recruited C fibres form a new afferent pathway** — the **rationale for intravesical capsaicin and RTX**.

Concept — Afferent recruitment is the unifying disease mechanism: *normal voiding runs on A δ tension receptors, whereas inflammation, SCI and obstruction unmask silent C fibres by lowering thresholds — which is why C-fibre toxins (capsaicin, RTX) and C-fibre signals (ATP, NGF, TRPV1) recur throughout the clinical sections.*

Efferent Pathways to the Bladder

- **Three pathways (Fig. 103.30):** pelvic parasympathetic excites bladder; hypogastric sympathetic inhibits bladder, excites neck and urethra; pudendal somatic supplies sphincter and pelvic floor striated muscle. Parasympathetic ACh contracts detrusor for voiding; sympathetic noradrenaline (NA) relaxes via β receptors during storage.

Terminal nerve fibres

- Fascicles → bundles with **dense innervation along the fascicle axis** from connective-tissue trunks → **widespread coordinated contraction**.
- Most detrusor nerves express **acetylcholinesterase and vesicular ACh transporter (VACHT)** — parasympathetic. **EFS responses need muscarinic + purinergic blockade** → ACh and ATP carry excitation; both released in muscle, persist after mucosal removal.
- Co-contents: **vasoactive intestinal polypeptide (VIP)**, **NOS**, **galanin**; suburothelial cholinergic nerves mostly also contain **neuropeptide Y (NPY)**, **tyrosine hydroxylase**, some **NOS**. **Trigone: VIP and NPY axons; noradrenergic fibres sparse in trigone, rare in detrusor, absent in urothelium**.
- **Parathyroid hormone-related protein (PTHrP)**: released by stretched DSM → **autocrine/paracrine relaxant**; suppresses spontaneous phasic contractions but barely affects carbachol/neural contractions → **stabilises filling without impairing voiding**. **PTH1R mainly in muscle; augmentation specimens are PTH1R-negative; detrusor tenfold more sensitive than muscularis mucosae**. Loss may cause **OAB or reduced capacity**; mechanism (possibly cAMP–PKA) unestablished.

Spinal ascending and descending influences: transmitters

- **Glutamate**: in primary afferent terminals, interneurons and medullary fibres; **generally excitatory** (vs **generally inhibitory glycine**), but effect depends on the postsynaptic neuron (**interposed inhibitory interneuron** → **net inhibition**). **NMDA receptors control lumbosacral polysynaptic reflexes** (Fig. 103.31); **glutamatergic input density ↓ with ageing**.
- **Glycine and γ -aminobutyric acid (GABA)**: projections **inhibit the micturition reflex** and glutamatergic neurons; **GABA receptor activation inhibits DO**. **Rectal distension inhibits bladder contractions** — abolished by intrathecal **strychnine** (glycine antagonist) + **bicuculline** (GABA-A antagonist) → **synergistic glycinergic/GABAergic rectovesical inhibition**.
- **Serotonin (5-hydroxytryptamine, 5-HT)**: raphe nuclei innervate lumbosacral autonomic nuclei; central 5-HT **suppresses voiding** (**inhibits parasympathetic drive**) and activates thoracic sympathetic neurons; **raphe stimulation inhibits reflex bladder activity (cat)**. **5-HT_{1A} and 5-HT₂ in sacral parasympathetic nucleus; species differ** — **5-HT_{1A} facilitates voiding in rats** (used to reverse diabetic dysfunction).
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor, SNRI): **↑ EUS neural activity and ↓ bladder activity centrally (cat)**; **↑ sneeze-induced mid-urethral continence reflex**, mainly via **α ₁-adrenoceptors** (rat); proposed for **SUI and UUI**.

- **Adrenergic:** descending catecholaminergic neurons in **upper medulla/pons**. Two spinal $\alpha 1$ mechanisms: **inhibitory (afferent processing)** and **excitatory ($\uparrow$ contraction amplitude via descending glutamatergic limb of the spinobulbospinal reflex)**. Rat lumbar cord: $\alpha 1A$ 70%, $\alpha 1B$ 30%, $\alpha 1D$ insignificant. Vascular $\alpha 1B$ blockade may $\uparrow$ pelvic blood flow (ageing, hypertension). $\beta 3$ -adrenoceptors occur peripherally and in the sacral cord; $\beta 3$ agonists (afferent + DSM effects) promising for OAB.
- **Purinergic:** ATP is co-released with NA and NPY (sympathetic) and with ACh (parasympathetic). Purinergic parasympathetic transmission is clear in rat, rabbit, guinea pig but minimal in normal human — emerges in DO and BOO. Seven P2X (ion channel) and eight P2Y (G-protein) subtypes; P2X1 dominates detrusor and bladder vascular muscle (rat, human) and $\uparrow$ in obstruction.
- **Fig. 103.41 — adrenergic terminal:** NE $\rightarrow \alpha 1$ contraction / β relaxation; $\alpha 2$ feedback inhibition; NPY contracts or inhibits ACh/NE release; ATP $\rightarrow$ P2 contraction; P1 prejunctional inhibition; ACh from adjacent terminals inhibits adrenergic firing via M1.

KEY POINTS: SPINAL CIRCUITRY NEUROTRANSMITTERS

- **Glutamate** is key in the spinal efferent micturition circuitry, is excitatory in **supraspinal** circuitry, and may mediate **DO after neural injury**.
- **Spinal $\alpha 1$ -noradrenergic system** modulates micturition: **inhibits bladder afferent input** and **facilitates the descending limb** to $\uparrow$ contractility.
- **5-HT, purines, glycine and GABA** selectively set the volume threshold in the sacral cord.
- Many substances modulate supraspinal circuits (Fig. 103.31); their **receptors are drug targets**.

Storage phase of the bladder

- **Low, constant filling pressure** below the voiding threshold (Fig. 103.33) depends on **intrinsic muscle/stroma properties, quiescent parasympathetic outflow, urothelial apical membrane changes and mediators**, and the **bladder sympathetic reflex** (negative feedback: closes outlet, inhibits contraction).

Reflex circuitry controlling continence and micturition

- **Central pathways = on-off switching circuits** (Fig. 103.32) with a **reciprocal bladder-outlet relationship**; low afferent activity $\rightarrow$ sphincter activation; **high afferent activity $\rightarrow$ switch flips** $\rightarrow$ bladder efferents fire, sphincter inhibited.
- Reflexes can chain: **bladder $\rightarrow$ sphincter guarding reflex $\rightarrow$ sphincter afferents $\rightarrow$ inhibition of bladder** (bladder-sphincter-bladder loop). **Sacral neuromodulation** probably works by activating these primitive reflexes.

TABLE 103.3 REFLEXES TO THE LOWER URINARY TRACT

Phase	Afferent pathway	Efferent pathways	Central pathway
Urine storage	Low-level vesical afferent activity (pelvic nerve)	EUS contraction (somatic); internal sphincter contraction (sympathetic); detrusor inhibition (sympathetic); ganglionic inhibition (sympathetic); sacral parasympathetic outflow inactive	Spinal reflexes
Micturition	High-level vesical afferent activity (pelvic nerve)	Inhibition of EUS; inhibition of sympathetic outflow; parasympathetic activation to bladder and to urethra	Spinobulbospinal reflex

- **What it tests: storage is spinal; voiding needs the pons.** The same pelvic afferents drive both — intensity decides which.
- **Sympathetic storage reflex** is organised in the **lumbosacral cord** and **survives thoracic transection** (Fig. 103.34: somatic cholinergic to EUS, adrenergic and cholinergic/nitrergic to urethral smooth muscle;

bulbospinal modulation) — **but is weak in humans: bilateral retroperitoneal lymph node dissection does not alter storage.**

- **Guarding reflex:** bladder afferents activate **pudendal motoneurons** (↑ sphincter EMG with filling); also triggered by urethral/perineal proprioceptive afferents. **Inhibition during micturition is supraspinal → weak or absent after chronic SCI → detrusor-sphincter dyssynergia (DSD).**

Sphincter-to-bladder reflexes

- **Pudendal somatic afferents inhibit voiding** — from **penis, vagina, rectum, perineum, urethral and anal sphincters** — by **suppressing sacral interneurons and directly inhibiting preganglionic neurons.**
- **EUS/pelvic floor contraction → proprioceptive afferents → central inhibition of micturition** (Fig. 103.35); in monkeys, anal sphincter stimulation inhibits voiding **even after chronic T4 paraplegia → partly spinal.**

Somatic-to-visceral reflexes

- **Sacral neuromodulation (SNM)** treats **refractory OAB and non-obstructive urinary retention (NOUR)**; proposed actions: **somatic afferent inhibition of sensory processing and restored brainstem autoregulation.**
- **Posterior tibial nerve stimulation (PTNS)** for OAB and NOUR; in cats **continuous tibial stimulation inhibits bladder activity beyond the stimulus**, involving **opioid and metabotropic glutamate** mechanisms.

Emptying phase of the bladder

- Switch to voiding is **involuntary (infant, neuropathic bladder: wall tension exceeds threshold) or voluntary.** Tension-receptor firing **reverses outflow:** sacral parasympathetic on, sympathetic and somatic off.
- Sequence: **urethral sphincter relaxes → seconds later detrusor contracts → pressure rises → flow.** **Urethral smooth muscle relaxation = parasympathetic NO + withdrawal of excitation; urethral flow reflexes aid emptying.** Parasympathetic organisation runs through **spinal and spinobulbospinal pathways via the pontine micturition centre (PMC).**
- **Fig. 103.33 (cystometrogram + sphincter EMG): infant (reflex) and adult (voluntary)** — sphincter relaxes **seconds before** contraction; stopping voiding → **sphincter EMG rises, bladder relaxes; paraplegic** — reciprocity lost, **uninhibited contractions with sphincter activity → bladder-sphincter dyssynergia**, poor emptying.

Urethra-to-bladder reflexes

- **Barrington (1931, 1941): urine flow or catheter stimulation of the urethra facilitates reflex bladder contraction** (promotes complete emptying). Two components:

BARRINGTON'S URETHRA-TO-BLADDER REFLEXES

Component	Afferent	Centre
First	Somatic — pudendal nerve	Supraspinal (PMC)
Second	Visceral — pelvic nerve	Spinal reflex

- **Low-frequency pudendal afferent stimulation (human) or deep perineal nerve (cat) initiates voiding.**
- **Rat:** stopping urethral perfusion or **intraurethral lidocaine ↓ reflex frequency**; **intraurethral NO donors (SNAP, nitroprusside) ↓ urethral pressure (about 30%) and reflex frequency (by about half or more) without changing amplitude** (Fig. 103.36); **intraurethral capsaicin first ↑ frequency, then blocks activity**

by 30 min (Fig. 103.37) → **urethral afferents modulate micturition** (link between **stress incontinence** and **detrusor instability**).

Concept — Reciprocity is the rule, intensity is the switch: low afferent traffic runs spinal guarding reflexes that close the outlet, high traffic trips the pontine switch that reverses every output at once, and losing the pontine link (SCI) leaves the spinal reflexes fighting each other — DSD.

Supraspinal Pathways

Pontine micturition centre and brainstem modulatory mechanisms

- **Barrington's cat lesions: micturition abolished by lesions at the inferior colliculus; facilitated by lesions anterior to it** (removal of inhibition). Animal pontine findings **translate to humans** (imaging; pontine lesions cause continence or retention problems).
- **Dorsal pontine tegmentum = PMC = Barrington nucleus = "M region"** (medial). **Discrete lesions → retention** (cat, rat); **human stroke or MS in the same region → retention**.
- **Stimulation of Barrington nucleus → bladder contraction + urethral sphincter relaxation**. Pontine units: **pre-contraction only; pre-contraction and sustained** (found in Barrington nucleus); **inhibited during contraction** (scattered).
- **Barrington nucleus does not project to Onuf's nucleus. L-region** (ventrolateral; **pontine continence/storage centre**, cat) projects to Onuf's nucleus — but **no Barrington-L-region connections**. Instead Barrington nucleus **inhibits Onuf's nucleus indirectly via excitatory projections to GABA premotor interneurons in the dorsal grey commissure** and **sacral intermediolateral inhibitory interneurons (glycine, GABA) → Barrington nucleus = command centre** for emptying.

Central circuitry by transneuronal tracing

- **Pseudorabies virus (PRV) retrograde tracing** from bladder or urethra labels **overlapping neurons with the same time course** → close central coordination.
- **First labelled (most direct to spinal efferents): ventral medullary raphe, parapyramidal reticular formation, A5, Barrington nucleus**. Later: **periaqueductal grey (PAG), hypothalamus, medial preoptic nucleus** (main inputs to Barrington), with **locus coeruleus (LC), cortex, red nucleus**. Distal colon labels **similarly** (shared pelvic control).
- PRV shows hierarchy, **not connectivity or function**. Human **voluntary control** needs **frontal cortex ↔ septal-preoptic hypothalamus** and **paracentral lobule ↔ brainstem**; cortical lesions **↑ bladder activity** (lost inhibition).
- **Fig. 103.38: interneurons relay to PMC and PAG; PMC receives PAG, lateral hypothalamus, medial preoptic input and projects to LC and lumbosacral parasympathetic neurons, and to dorsal grey commissure premotor neurons → Onuf's nucleus; pontine continence centre (PCC) = L-region**.

Neurotransmitters and modulators within brainstem networks

- **Glutamate** is the principal transmitter of Barrington neurons to preganglionic parasympathetic neurons (**NMDA and non-NMDA**). Fig. 103.31: glutamate acts on **NMDA and AMPA** receptors in ascending and descending limbs and sphincter pathway; modulators **GABA, enkephalins, ACh, dopamine (ACh and dopamine both excitatory and inhibitory)**.
- **Corticotropin-releasing factor (CRF)** in Barrington neurons is **inhibitory** at the spinal preganglionic level (**CRF antagonist enhances evoked contraction**).
- **Serotonin: inhibitory to detrusor, excitatory to urethral sphincter**, at dorsal horn and motoneuron levels; **presynaptic 5-HT_{1A} on afferent terminals ↓ transmitter release**.
- **Fig. 103.31 (cat): normal micturition = supraspinal PMC reflex triggered by Aδ tension receptors; cortical-diencephalic modulation; after cord transection micturition is first blocked, then a C-fibre spinal**

reflex emerges; **ice water activates it in SCI**; **capsaicin (20–30 mg/kg s.c.) blocks C-fibre reflexes in chronic spinal cats but not intact cats**; **intravesical capsaicin suppresses instability and cold reflexes in neurogenic bladder.**

Human brain imaging studies

- **Single-photon emission computed tomography (SPECT), positron emission tomography (PET), functional MRI (fMRI)** — indirect blood-flow measures. **PET: long-lasting states; fMRI: fast events; SPECT: poor spatial and temporal resolution.**
- **Fig. 103.39: storage activates thalamus, insula, prefrontal cortex, anterior cingulate, PAG, pons, medulla, supplementary motor area (SMA).**

Cerebral control of voiding

- **Voiding = long-loop spinobulbospinal reflex:** filling afferents → sacral cord → **PAG (± PMC)**; above trigger → **PMC efferents** → indirect inhibition of Onuf's nucleus (sphincter relaxation) + excitation of detrusor. A switch — "off" (storage) or "on" (voiding); without higher control, involuntary emptying at a critical volume.
- **White-matter damage causing permanent incontinence** interrupts the **medial frontal → brainstem (direct or via thalamus)** pathway that **tonically inhibits voiding during storage.**
- **Storage/withholding activates right inferior frontal or dorsolateral prefrontal cortex** (right-sided predominance); little medial frontal activation — **imaging shows grey matter; lesions cut white-matter links.**
- **MS with DSD: anterior thalamic radiation (ATR) involvement; transcranial magnetic stimulation (TMS) of these areas ↓ voiding symptoms.**

Additional regions

- **PET during voiding: dorsolateral prefrontal cortex, anterior cingulate gyrus (ACG), hypothalamus/preoptic area, pons, PAG;** also inferior frontal gyrus, postcentral and superior frontal gyri, thalamus, insula, globus pallidus, cerebellar vermis.
- **ACG responds to filling/withholding; dorsal ACG and SMA over-respond in urge incontinence** (urgency without contraction). **Urgency** = compelling, hard-to-defer desire to void with **fear of leakage** — a **homeostatic and social emotion.** Rats activate cingulate cortex on filling (homology and "rat urgency" unknown).
- **Human PMC:** posterior fossa tumours, **discrete pontine lesions, childhood brainstem gliomas, vascular lesions → retention;** MRI localises to **dorsolateral pons** (pontine reticular nucleus/formation, **near medial parabrachial nucleus and LC**); **bladder symptoms overlooked** amid disturbed consciousness and respiration.
- Also parietal/frontoparietal, **precuneus, posterior cingulate, hippocampus, amygdala, cerebellum, basal ganglia (striatum, putamen).** Dopamine tonically inhibits the PMC; lost in Parkinson disease (→ overactivity).

Model of brain–bladder control and normal continence mechanism

- Circuits answer: **Is the bladder full enough? Is voiding safe and socially appropriate?** → **limbic (emotion, safety) and cortical (propriety, decision) control of the brainstem switch = PAG + PMC;** PMC is the **final efferent brain nucleus;** PAG receives medial and orbital prefrontal input.
- Storage sequence: **filling afferents → PAG (switch not triggered) → thalamus → insula (desire to void) → lateral and medial prefrontal cortex (decision, social appraisal) → if no void planned, medial frontal → brainstem pathway (direct or ATR) tonically suppresses the reflex = normal continence mechanism;** **medial prefrontal cortex–PAG (mPFC–PAG) white-matter damage → incontinence.** Usually no conscious bladder awareness.

- **Fig. 103.40:** normal (sensation-based) mechanism vs **backup mechanism = urgency**, via **dorsal ACG/SMA, L-region (pontine storage centre)** or sympathetic outflow; a third circuit monitors safety without conscious sensation.

Concept — The pons switches, the forebrain decides: PAG + PMC form a binary reflex switch, and continence is the medial prefrontal cortex tonically holding it off — so white-matter or frontal lesions produce incontinence, pontine lesions produce retention, and urgency is the forebrain's emergency backup.

PHARMACOLOGY

Muscarinic Mechanisms

- Human detrusor strips contract to muscarinic agonists and cholinergic nerve stimulation — abolished by atropine. Five subtypes (M1–M5) by cloning; four pharmacologically defined.
- Human bladder: M1, M2, M3 by binding; all M1–M5 mRNA present. M2 predominates numerically, M3 mediates contraction.
- M3 → PLC → IP₃ → Ca²⁺ release; L-type entry also important (nifedipine strongly blocks carbachol contraction; PLC or SOCE inhibitors little effect) — but PLC inhibitors do suppress in other rat studies, and Ca²⁺ rises without depolarisation in human cells. Guinea pig: both entry and store release.
- M2 enhances M3 contraction by: (1) inhibiting adenylyl cyclase (opposing sympathetic β relaxation); (2) inactivating K⁺ channels; (3) activating non-specific cation channels.
- ROK inhibitor Y-27632 suppresses carbachol contraction (rat, human) → muscarinic activation recruits Rho kinase → calcium sensitisation; mainly M3, possibly M2 (Y-27632 ↑ darifenacin affinity; darifenacin ≈ 30-fold M3 > M2).
- Shift from M3 to M2 in obstructed or denervated hypertrophied rat bladders and human neurogenic bladder.
- M3 knockout: salivation, pupil constriction and detrusor contraction depend on M3 (gut/reproductive redundant); male knockouts: distended bladder, larger capacity (sex difference) → model for DO.
- Oxybutynin: non-selective antimuscarinic + direct smooth muscle relaxation — relevant only intravesically. Botulinum toxin A in paediatric neurogenic DO (NDO) ↓ muscular muscarinic receptor expression.
- Metabolites matter: desethyloxybutynin causes more dry mouth than oxybutynin → controlled-release oxybutynin keeps efficacy with fewer side effects. Tolterodine (cat) and solifenacin (rat) act less on salivary than bladder receptors. Site and speed of metabolism shape efficacy and side effects.

TABLE 103.4 DRUGS WITH BLADDER ACTION

Class	Examples	Target / mechanism	Intended effect
Anticholinergics	Atropine, glycopyrrolate, oxybutynin, propantheline, tolterodine	Muscarinic antagonism	↓ Filling pressure; ↓ unstable contractions
Smooth muscle relaxants	Dicyclomine, flavoxate	Direct relaxation (most also anticholinergic)	↓ Filling pressure; ↓ unstable contractions
Calcium antagonists	Diltiazem, nifedipine, verapamil	↓ Ca ²⁺ entry during AP	↓ Spike magnitude of unstable contractions
K ⁺ channel openers	Cromakalim, pinacidil	Hyperpolarise membrane	↓ Myogenic initiation of unstable contractions

Class	Examples	Target / mechanism	Intended effect
PG synthesis inhibitors	Flurbiprofen	↓ PGs (which ↑ tone and spontaneous activity)	Relaxation during filling; ↓ spontaneous activity
β-agonists	Isoproterenol, terbutaline	β → bladder body relaxation	↓ Filling pressure
Tricyclic antidepressants	Amitriptyline, imipramine	Anticholinergic + direct relaxant + NA reuptake inhibition	Storage
α-agonists	Ephedrine, phenylpropanolamine, midodrine, pseudoephedrine	α → urethral contraction	↑ Urethral tone and closure pressure
Afferent nerve inhibitors	DMSO, capsaicin, resiniferatoxin	↓ Sensory input	↑ Capacity; ↓ DO
Oestrogen	Estradiol (vaginal or oral)	↑ Urothelial/mucosal thickness (seal); ↑ urethral adrenergic effect; ↑ blood flow	↓ Incontinence

- **What it tests:** every storage drug acts at one of **four points** — **muscarinic receptor, muscle membrane** (Ca^{2+} in, K^{+} out), **afferent nerve, or outlet** (α , oestrogen). **Trap:** α -agonists and oestrogen act on the outlet; all others on the bladder. **Tricyclics hit three mechanisms at once.**

Muscarinic Selectivity

ANTIMUSCARINIC SELECTIVITY

Drug	Receptor subtype selectivity	Tissue (bladder vs salivary) selectivity
Darifenacin, vamicamide	Relatively M3-selective (darifenacin ≈ 30-fold)	Not tissue-selective (salivary glands also M3)
Tolterodine	Not M3-selective	Bladder-selective
Solifenacin	10-fold M3 > M2 (similar to oxybutynin)	Bladder-selective
Oxybutynin	~10-fold M3 > M2; non-specific	Not selective; metabolite drives dry mouth

- Tissue selectivity matters more than subtype selectivity; a truly bladder-selective antimuscarinic is the "Holy Grail" of OAB therapy.
- Urothelium releases non-neuronal ACh and ATP; ACh receptor stimulation releases ATP, blockade reduces it → antimuscarinics may act partly via the urothelium/afferents.

Cognitive Impairment With Antimuscarinic Agents

- **CNS M1 blockade** (lipophilic, blood-brain barrier-crossing agents) → **cognitive impairment**, especially elderly and comorbid. **Additive anticholinergic burden** (antihistamines, antispasmodics, muscle relaxants) → **confusion, mild cognitive decline, delirium**. **Withdrawal improves central effects within weeks**. **Caution in the frail elderly, more so with pre-existing dementia.**

KEY POINTS: MUSCARINIC MECHANISMS

- At least five subtypes; M1, M2, M3 found pharmacologically in human bladder.
- M3 → L-type Ca^{2+} influx + PLC/IP3 Ca^{2+} release, both contributing to contraction.
- Shift from M3 to M2 mediation reported in **neurogenic bladder**.

Adrenergic Mechanisms

β-Adrenergic Receptors

- **β2 and β3 in human detrusor → direct relaxation via adenylyl cyclase → ↑ cAMP.** Relaxation not blocked by β1/β2 antagonists but blocked by β3 antagonists → **β3 mediates it; β3 is the most highly expressed adrenoceptor mRNA in human bladder.**
- **Mirabegron (β3 agonist) is approved for OAB with UUI**, including for antimuscarinic contraindication/intolerance; **mirabegron + solifenacin combination is effective.** Acts on **afferents** too: in rodents ↑ compliance, ↓ non-micturition contractions and afferent activation (shortened intervoid interval also reported).
- **PDE inhibitors also ↑ cAMP/cGMP;** PDEs differ in substrate, species, tissue; search for **bladder- vs penis-specific isoforms.** PDE4 degrades cAMP in guinea pig phasic cells; **PDE4 inhibitors ↓ DO in rat BOO.** PDE1 and PDE5 inhibitors used clinically for storage/voiding disorders.

α-Adrenergic Receptors

- **Not prominent in normal bladder**, but in **BOO-associated DO**, α-density (especially α1D) may rise enough to **convert NA response from relaxation to contraction.**
- **Rat obstruction: α1D share rises from 25% to 70%;** frequency suppressed by **tamsulosin (α1D + α1A)** but not by **5-methyl-urapidil (α1A only);** α1D knockouts: **larger capacity and voided volume.**
- **Human: α1D already predominates in normal bladder**, total α mRNA is **low vs β3** and not ↑ by obstruction → human role unproven.
- **Urethra is where α matters: α1 and α2 present; α1A is the main urethral smooth muscle subtype;** α-agonists contract isolated urethra; **hypogastric stimulation raises urethral pressure, blocked by α1 antagonists** → rationale for α-agonists to ↑ outlet resistance.
- **α-antagonists aid emptying in functional outlet resistance (BPH).** Selective α1A blockers (e.g. RS-17053) relax prostate and ↑ flow but do not improve LUTS scores; antagonists with α1D activity improve storage symptoms → α1D (bladder or spinal) drives storage symptoms of BOO.
- **α2: antagonists ↑ urethral NA release (presynaptic) without altering contractility;** human urethra lacks postjunctional α2; prejunctional α2 gives **feedback inhibition.** Hypogastric nerves inhibit vesical ganglionic cholinergic transmission via α2; β-agonists facilitate it (cat).

ADRENOCEPTOR SUBTYPES IN THE LOWER URINARY TRACT

Subtype	Site	Effect	Clinical relevance
β3	Detrusor (dominant adrenoceptor); sacral cord	Relaxation via cAMP	Mirabegron for OAB
α1A	Urethra, prostate (dominant)	Contraction	α-agonists ↑ closure; α1A block ↑ flow, not LUTS
α1D	Bladder (human normal; ↑ in rat BOO); spinal	Contraction / storage symptoms	α1D blockade improves storage symptoms
α1B	Blood vessels; spinal (30%)	Vasoconstriction	Blockade ↑ pelvic blood flow
α2	Prejunctional urethra; vesical ganglia	↓ NA release; ↓ ganglionic transmission	No postjunctional urethral effect

KEY POINTS: ADRENERGIC MECHANISMS

- **β-relaxation = adenylyl cyclase → cAMP.**
- **β3 is the most highly expressed adrenoceptor; β3 agonists (mirabegron) are now approved for OAB.**
- **Human bladder: α1D predominates normally; α mRNA low vs β3 and not ↑ by obstruction.**

- **α -receptors govern urethral tone; α 1A dominates prostate and urethra; all three α 1 subtypes in blood vessels.**

Urethral Tone in Females

- Regional study of the female urethra: **sympathetic α 1 tone maintains resting urethral tonus. α 1A/1L agonism $\downarrow$ incontinence episodes in mild-moderate SUI** (small placebo-controlled study) $\rightarrow$ possible urethra-specific adrenergic agonists.

Concept — Receptor location decides drug action: *the detrusor relaxes through β 3 and contracts through M3 (with M2 as amplifier), whereas the outlet closes through α 1A — so bladder drugs target β 3/M3, outlet drugs α 1, and α 1D sits between as the storage-symptom receptor of obstruction.*

Afferent Neuropeptides

- LUT afferents contain **SP, calcitonin gene-related peptide (CGRP), pituitary adenylate cyclase-activating polypeptide (PACAP), leucine enkephalin, CRF, VIP, plus growth-associated protein-43 (GAP43), NOS, glutamate, aspartate** — in DRG, peripheral endings and spinal terminals.
- **>70% of rat bladder DRG neurons contain multiple peptides** — CGRP, SP, PACAP commonest; VIP common in cat.
- Released from **capsaicin-sensitive C fibres by noxious stimuli** $\rightarrow$ **plasma extravasation, vasodilation, altered DSM activity** (neurogenic inflammation); also **spinal transmitters**.

Tachykinins

- Shared C-terminal **Phe-Xaa-Gly-Leu-Met-NH₂**; **SP, neurokinin A (NKA), neurokinin B (NKB)**; mainly in **unmyelinated C-fibre terminals**; **three seven-transmembrane G-protein receptors**, all present in bladder.

TABLE 103.5 TACHYKININS AND TACHYKININ RECEPTORS (WITH BLADDER ACTIONS FROM TEXT)

Tachykinin	Preferred receptor	Bladder actions
Substance P	NK1	Vessels: plasma extravasation, vasodilation; spinal NK1 carries non-nociceptive afferent input
Neurokinin A	NK2	Detrusor contraction; NK2 autoreceptors sensitise afferents (high urinary K ⁺ , distension)
Neurokinin B	NK3	$\uparrow$ C-fibre excitability; spinal NK3 inhibits micturition via opioid mechanism

- **Intrathecal or centrally acting NK1 antagonists $\uparrow$ capacity without changing voiding pressure**; NK2, NK3 or peripheral NK1 antagonists ineffective on normal voiding. **NK1 antagonists $\downarrow$ DO** (cystitis, capsaicin, L-dopa); **NK2 antagonist $\downarrow$ capsaicin DO only. TAK-637 $\uparrow$ volume threshold.**
- **Aprepitant (NK1 antagonist) $\downarrow$ micturitions and urgency vs placebo at 8 weeks in women with idiopathic OAB.**
- **Conclusion: non-nociceptive afferent input $\rightarrow$ spinal NK1; nociceptive input $\rightarrow$ NK1, NK2 and NK3.**
- **NK2 autofeedback:** high K⁺ or distension $\rightarrow$ **NKA release $\rightarrow$ NK2 autoreceptors $\rightarrow$ sensitisation; SR 48968 (NK2 blocker) $\downarrow$ mechanoreceptor sensitivity and blocks K⁺ sensitisation. DRG: NK2 excites, NK3 inhibits (PKC-mediated Ca²⁺ channel modulation); NK2 $\rightarrow$ PKC $\rightarrow$ TRPV1 phosphorylation $\rightarrow$ $\uparrow$ capsaicin currents.**

Prostanoids

- **PGs and thromboxanes** from arachidonic acid via **COX-1 and COX-2**; roles in **contractility, inflammation, neurotransmission**. Human mucosa: **PGI₂, PGE₂, PGF₂ α , thromboxane A₂**. **Contractile potency: PGF₂ α > PGE > PGE₂.**

- Receptors: **DP, EP, FP, IP, TP** (for PGD₂, PGE₂, PGF₂ α , PGI₂, TXA₂); **EP1-EP4**. **Slow onset → modulatory**: ↑ neural transmitter release, **inhibit acetylcholinesterase** → augment cholinergic contractions.
- **Intravesical PGE₂ to aid voiding: mixed results** (helped some retention/neurogenic patients; not others) but it **causes urgency and involuntary contractions**; **indomethacin ↓ DO**.

Endothelins

- **21-amino-acid peptides ET-1, ET-2, ET-3** (separate genes); receptors **ETA (ET-1 = ET-2 > ET-3) and ETB (all equal)**. **ET-1 → prolonged bladder contraction**; ET immunoreactivity in almost all bladder cells → **tone, blood flow, remodelling**.
- **BOO (rabbit)**: ↑ ET-1, ETA (DSM, urothelium) and ETB (DSM); **ET-converting enzyme inhibitor ↑ voiding efficiency and ↓ DO**; ETA antagonist (YM598) ↓ DO → ET may drive **hyperplasia and DO in BPH-related BOO**.
- **Afferent role**: ETA on C fibres → DO; spinal ETA → **inhibits micturition via opioids**; SCI: ↑ bladder ET-1; ETA antagonist (ABT-627) ↓ C-fibre DO.

Sex Steroids

- Symptoms vary through the **menstrual cycle**. **Sex steroids do not contract bladder directly; they modulate receptors and growth**. **Oestrogen receptors in the female trigone**.
- **Conflicting animal data**: oestrogen ↑ responsiveness to α , cholinergic and purinergic agonists (rabbit) vs ↓ adrenergic/muscarinic density and ↓ contractions (rat).
- **Oestrogen ↑ urethral adrenergic receptors**; **unmasks α -contraction in ovariectomised rabbits** → **oestrogen + α -agonist combined for SUI, efficacy questioned**. Oestrogen acts via **adrenergic receptors, vasculature and urothelium**. **Progesterone ↑ electrical and cholinergic contractions**; both induce NOS (possible DO relief).
- **Clinically**: oestrogen alone **disappointing for SUI or UUI**, and **systemic oestrogen may ↑ UI in postmenopausal women**.
- **Androgens (male rat)**: alter synapses and **size of androgen-sensitive bulbocavernosus/levator ani motoneurons**; **testosterone enlarges major pelvic ganglion neurons**.

CLINICAL RELEVANCE

- **Ideal treatment reverses the cause** (neural injury, obstruction, infection); **idiopathic LUTD (OAB, IC/PBS, UAB) is treated empirically** for symptom relief.
- Many idiopathic conditions reflect **augmented sensory input** → "**afferent neurourology**". **Idiopathic NOUR (underactive detrusor)** may be **efferent failure**, worsened by ↓ **motor-sensory** feedback.
- **Closed-loop optogenetics (mouse)**: cyclophosphamide DO detected by pressure sensors → light activates **archaerhodopsin 3.0** in afferents (herpes simplex vector) → ↓ afferent activity → ↓ DO.
- **Overarching paradigm: C-fibre activation via neurotrophins (NGF)** made by DSM, **temporally linked to DO** (Fig. 103.42: SCI, obstruction, inflammation, denervation, spontaneously hypertensive rat → ↑ neurotrophic factors → hyperexcitable C fibres → **spinal positive-feedback loops**; hypertensive rat also ↑ α 1 excitation). **Local anaesthetics, intravesical neurotoxins and bladder neck/prostate afferent destruction ↓ urgency, frequency, UUI**.
- **Ice-water (spinal C-fibre) reflex appears in neurogenic bladder and BOO** → shared plasticity; **BP-LUTS association in BPH** links sympathetic tone to symptoms.

Mechanisms of Idiopathic Detrusor Overactivity and Overactive Bladder

- **Classic model** = urodynamic DO, yet DO found in only 11% of an OAB cohort.
- **Non-injury DO models**: spontaneously hypertensive rat; serotonin reuptake transporter knockout; BK channel knockout; urothelial NGF knock-in; urothelial $\beta 1$ -integrin knockout. Other knockouts (M1–M5, P2X2, P2X3, TRPV1) study LUT function but not OAB. Theme: growth-factor-driven neural and muscle plasticity.
- **NGF as biomarker**: $\uparrow$ by DSM after BOO and diuresis; urinary NGF studied for OAB (source uncertain — unlikely stromal, as it must cross lamina propria and urothelium). Urothelial NGF overexpression $\rightarrow$ frequency, pelvic pain, lamina propria neuronal hypertrophy. Paradox: no $\uparrow$ urothelial NGF in DO; less in frequency $\rightarrow$ role unresolved.
- **Cultured OAB urothelium**: $\uparrow$ polyamine signalling $\rightarrow$ blocks urothelial BK channels; $\uparrow$ TRPV1 signalling.
- **Urethral afferent function** $\downarrow$ in OAB/UUI (current perception threshold testing) and unchanged after successful SNM — counterintuitive.
- **DSM**: BK knockout $\rightarrow$ OAB phenotype; DO myocytes (IDO, NDO) show $\uparrow$ resting $[Ca^{2+}]_i$ and spontaneous rises via L- and T-type channels, greater in NDO.
- **Suburothelial myofibroblasts in IDO/NDO**: $\uparrow$ Cx43, not reversed by onabotulinumtoxinA (OBTX). Cytokines (IL-4, IL-6, IL-10, TNF- α , TGF- $\beta 1$) $\uparrow$ Cx43 in myofibroblasts and myocytes; NGF $\uparrow$ Cx43 coupling in neuronal cells $\rightarrow$ $\uparrow$ connectivity in OAB and IC/PBS.

Stress Urinary Incontinence

- **Integral theory** (Petros and Ulmsten, 1993) $\rightarrow$ mid-urethral sling (MUS), the surgical standard. But DeLancey: intrinsic sphincter function matters more than anatomical support.
- **Childbirth injury models** (vaginal distension, pudendal injury): stem cell homing chemokines $\uparrow$ (CCL7, CXCR4, CXCL12, CD195, CD193; monocyte chemoattractant protein-3, MCP-3) $\rightarrow$ maximise post-partum urethral healing to prevent SUI.
- **Biomarker**: higher urinary N-telopeptide (NTx; bone turnover) $\rightarrow$ higher MUS failure. MUS failure at 1 year about 23% (definition-dependent).
- **Duloxetine** (SNRI): $\uparrow$ Onuf's nucleus output $\rightarrow$ $\uparrow$ striated sphincter activity $\rightarrow$ $\downarrow$ SUI; beats placebo in mild-moderate SUI; approved in Europe, not USA; small benefit in post-prostatectomy SUI.

Spinal Cord Injury and Neurogenic Detrusor Overactivity

- **Suprasacral cord damage** $\rightarrow$ DO. Spinal shock (retention, a few weeks) $\rightarrow$ NDO via a new spinal micturition reflex from synaptic reorganisation; previously unresponsive afferents become mechanosensitive.

A δ VS C BLADDER AFFERENTS IN THE MICTURITION REFLEX (FIGS 103.24, 103.25)

Feature	A δ fibres	C fibres
Myelination	Lightly myelinated	Unmyelinated
Proportion	About 30% (some species)	Majority
Normal behaviour	Low-threshold tension receptors; drive normal reflex	Silent to gradual distension (cat; rat may fire at low pressure)
Reflex	Spinobulbospinal (via PMC)	Spinal reflex after SCI, neurological disease, ageing, ?inflammation
Capsaicin/RTX	Resistant	Sensitive — blocked

- **Capsaicin-sensitive C fibres** implicated in DO of SCI and MS; $\uparrow$ TRPV1 and P2X3 in suburothelial nerves and TRPV1 in basal urothelium, reversed by intravesical capsaicin or RTX.

- **DRG plasticity after SCI:** cell enlargement, ↑ excitability; ↑ tetrodotoxin (TTX)-sensitive Na⁺ channels, ↓ TTX-resistant Na⁺ channels and low-threshold A-type K⁺ channels.
- **NGF drives it:** chronic NGF → overactivity and ↑ afferent firing; bladder and spinal NGF ↑ after SCI; intrathecal anti-NGF ↓ DO and DSD and blocks autonomic dysreflexia → NGF a target for DO and DSD.
- C-fibre spinal reflex = **positive feedback, unresponsive to voluntary control**. Ice-water test reliable in research but **not sensitive or specific clinically**.
- **NDO:** ↓ BK channel expression and activity (BK drives phases 2–3 repolarisation) → ↑ contractions; possibly **adaptive** (↑ contractility to aid emptying without normal innervation).

KEY POINTS: NEUROGENIC BLADDER

- Normal micturition: spinobulbospinal reflex via lightly myelinated Aδ afferents.
- C fibres are normally insensitive to gradual distension.
- After SCI a capsaicin-sensitive C-fibre spinal reflex develops and may cause DO.
- Intravesical capsaicin or RTX improves a subpopulation of NDO patients and ↓ TRPV1 in nerves and urothelium.
- Ice-water test = research tool, **not clinical practice**.

Nocturia

- Especially isolated nocturia has distinct causes: **global polyuria, nocturnal polyuria, reduced capacity, sleep disorders, heart failure, circadian clock disorders**.
- Mouse bladder Cx43 is circadian: inactive (light) phase ↓ Cx43 → ↑ capacity; active (dark) phase ↑ Cx43 → ↓ capacity. Cry clock-gene knockout abolishes both oscillations → clock failure may cause nocturia and nocturnal enuresis.

Bladder Outlet Obstruction

- Symptoms of urethral obstruction are mostly bladder-generated: detrusor hypertrophy and DO.
- Rat partial obstruction: bladder about 15× heavier, same shape; surface sixfold, muscle threefold thicker; no mitoses — massive cell hypertrophy.
- Muscle mechanisms: denervation supersensitivity to cholinergic agonists; contractile protein change — SM-A expression induced, ↑ force but ↓ cross-bridge cycling rate, altered SM1 : SM2.
- Myogenic theory (Brading and Turner): all DO shares a DSM change predisposing to unstable contraction; pig obstruction DO occurs **without a micturition reflex**. IDO detrusor shows ↑ electrical coupling, patchy denervation, K⁺ supersensitivity.
- Partial denervation (↓ ACh release and nerve fibres 3–6 months after ligation) → muscarinic supersensitivity, ↑ purinergic (P2X1) responses, ↑ Cx43 (also in NDO and urgency) → coordinated whole-bladder myogenic contraction. Interstitial cells ↑ in BOO; c-KIT inhibitors ↓ spontaneous contractions.
- Neural plasticity: enhanced spinal reflex (rat); ice-water reflex in obstructed humans; spinal GAP43 ↑ (axonal sprouting) in sacral parasympathetic nucleus; afferent and efferent neuron hypertrophy. New circuits may not reverse after relief (persistent frequency). Muscle and neural changes co-exist.
- NGF: ↑ in obstructed bladders (animals, humans), precedes neuronal enlargement and frequency; NGF autoantibodies prevent plasticity and frequency; stays ↑ with persistent frequency after relief → cause and effect. fMRI: supraspinal changes in men failing symptom relief after BOO reduction.
- Urinary NGF: controls 0.5 pg/mL, BOO without OAB 1, BOO + OAB 41, BOO + DO 50. NGF ↑ after MUS — consistent with MUS ↑ Pdet at Qmax (a urodynamic obstruction).

KEY POINTS: BLADDER OUTLET OBSTRUCTION

- Obstruction-induced DO attributed to denervation supersensitivity; also contractile protein changes.
- ↑ Receptor-mediated contractility and cell-cell interaction.
- BOO enhances a spinal reflex (rat).
- Bladder tissue and urinary NGF ↑ in obstruction.

Concept — Obstruction remodels both ends of the reflex: NGF from the strained detrusor enlarges afferents and builds new spinal circuits while the partly denervated, better-coupled muscle becomes supersensitive — which is why frequency can persist after the obstruction is relieved.

Underactive Bladder

- UAB = LUTS suggesting ↓ contractility (weak stream, hesitancy, interruption, straining), often with frequency, urgency, nocturia; may be secondary to BOO (prior sling, BPH) → non-specific, not a diagnosis of cause.
- Detrusor underactivity = urodynamic ↓ contractility; poorly correlated with LUTS; afferent loss may cause it (↓ sensory input → ↓ contraction), not just efferent failure.
- Post-void residual (PVR): expert panel — ≥300 mL before further evaluation; also evaluate if LUTS moderately–severely impair quality of life or risk factors (hydronephrosis, renal insufficiency, culture-proven UTI). PVR <300 mL does not prove normal contractility.

Bladder Pain Syndrome and Interstitial Cystitis

- Pain attributed to the bladder + frequency and urgency; no defined cause.

BPS/IC DIAGNOSIS — EUROPE VS UNITED STATES

Feature	Europe	United States
Basis	Biopsy showing overt inflammation	Symptoms (chronic pain, frequency, urgency; no UTI)
Histology	Oedema, vasodilation, nerve proliferation, mast cells	Biopsy rare; rarely deep
Test	Biopsy	Hydrodistension — glomerulations

- Glomerulations (petechiae) after hydrodistension: first described by Messing and Stamey (1978); specificity questioned.
- Inflammatory? — debated. Cyclophosphamide haemorrhagic cystitis is used as a model but is unrelated to BPS/IC; relevance needs clinical validation.
- Chemical cystitis → sensitised mechanosensitive afferents and recruited silent C fibres; PGE₂, 5-HT, histamine, bradykinin, adenosine, NGF → hyperexcitable C fibres. Chronic irritation ↑ afferent NOS, GAP43, PACAP, SP, protease-activated receptors; peptidergic sprouting in mucosa and detrusor, close to efferent varicosities.
- Spinal changes: ↑ c-FOS; ↑ GFRα1 (smaller GFRα3, none GFRα2); phospho-ERK (extracellular signal-regulated kinase) ↑ by noxious stimuli normally and by innocuous stimuli after inflammation; intrathecal ERK inhibition (PD98059) ↓ reflex activity and c-FOS only in inflamed bladders → spinal ERK mediates inflammatory pain and overactivity.
- Afferent neurons in cystitis (L6–S1 DRG): normally capsaicin-sensitive, high-threshold TTX-resistant spikes; cystitis → lower threshold, tonic (not phasic) firing, larger soma, ↑ capacitance, ↓ A-type K⁺ (KA) current via ↓ Kv1.4 α-subunit; feline IC: larger, hyperexcitable neurons (↓ low-threshold K⁺ currents); Aδ afferents also more pressure-sensitive.

- **TRPV1 in IC:** feline IC DRG → **larger, slower-desensitising capsaicin responses, reversed by PKC inhibition**; TRPV1 knockouts lack cyclophosphamide/acrolein overactivity and referred mechanical hyperalgesia; oral TRPV1 antagonist GRC-6211 ↓ overactivity. Capsaicin/RTX helped IC pain, but the randomised RTX trial was negative.
- **Cross-sensitisation:** rectal irritant → **bladder afferent sensitivity, C-fibre involvement, mast cell activation** (BPS/IC + IBS).
- **NGF:** expressed by urothelium, muscle and **mast cells (triggers degranulation, proliferation)**; cyclophosphamide ↑ **NGF, brain-derived neurotrophic factor (BDNF), ciliary neurotrophic factor (CNTF), TrkA/TrkB phosphorylation, phospho-CREB** (cAMP-responsive element-binding protein) — **reduced by RTX**. Urinary NGF, neurotrophin-3, glial cell line-derived neurotrophic factor (GDNF) detected in BPS/IC; ↑ NGF in IC biopsies. Exogenous NGF → **nociception and overactivity**; NGF sequestration (TrkA-IgG, REN1820) ↓ **hyperalgesia and cystitis overactivity**; urothelial NGF mice: frequency, pain, hyperinnervation, ↑ mast cells. But tanezumab (anti-NGF antibody) failed in BPS/IC — perhaps **phenotyping** (enrol only high-NGF patients).
- **Purinergic:** ↑ urothelial ATP release (patients, cats); **P2X antagonists (PPADS, A-317491) ↓ non-voiding contractions and frequency in cystitis**; ↑ **afferent firing mimicked by purinergic agonists, blocked by P2X antagonists**; P2X3 (thoracolumbar) and P2X2/3 (lumbosacral) currents ↑. Human IC urothelium: **stretch → ↑ ATP and ↑ P2X3**. P2X3 antagonist AF-219 improved pain, urgency and global symptoms in a proof-of-concept trial.
- **Water avoidance stress (WAS)** (rat on a platform over water) → **visceral (incl. bladder) hyperalgesia, ↑ mast cell activity** (blocked by **melatonin, montelukast**) → brain mechanisms (fMRI).
- **Antiproliferative factor (APF):** urinary **sialated glycopeptide (nonapeptide)** homologous to **frizzled 8 (Wnt pathway)**; receptor **CKAP4/p63** (links ER to microtubules; receptor for tissue plasminogen activator and surfactant protein A; substrate of tumour-suppressor **palmitoyl acyltransferase DHHC2**) → antiproliferative on urothelium; **intravesical APF reproduces human urothelial changes in mice**.

KEY POINTS: BLADDER PAIN SYNDROME AND INTERSTITIAL CYSTITIS

- **Histology (often):** oedema, vasodilation, nerve proliferation, mast cell infiltration.
- **NGF links inflammation to altered pain signalling**; made by urothelium, muscle and mast cells; **activates mast cells**.
- **NGF sequestration ↓ cyclophosphamide cystitis overactivity**.
- **Chronic inflammation alters C-fibre function and chemosensitive receptors (TRPV1)**.

Concept — BPS/IC is modelled as afferent hyperexcitability: *inflammation, stress or neighbouring-organ irritation raise NGF, ATP and TRPV1 activity and lower C-fibre thresholds — yet single-target trials (anti-NGF, RTX) have failed, pointing to heterogeneous patients rather than a wrong mechanism.*

Aging

- **LUTS common in the elderly:** frequency, urgency, UUI, poor emptying; described changes include ↓ capacity, ↑ sensation, DO.
- **Community-dwelling women:** detrusor contractility, bladder sensation and urethral pressure decline with age; capacity does not fall with age but is smaller with DO → ageing = hypofunction of bladder and urethra. Pelvic ischaemia (vasculopathy) → LUTS, DO.
- **Ageing is the largest risk factor:** cellular senescence, oxidative stress → ↓ perfusion, visceral hyposensitivity, abnormal urodynamics. **Targeting Aging with Metformin (TAME) trial** targets ageing

pathways; **metformin extends murine lifespan**; **caloric restriction prevents age-related bladder dysfunction in vitro**.

- **Aged rats:** ↑ voided volume and **higher pressure threshold**; ↓ **afferent sensitivity to volume (not pressure)**; ↓ maximal evoked pressure. **Aged mice:** **normal contractility**, ↓ **afferent signalling**.
- **Human:** linear ↓ **acetylcholinesterase-positive nerves** (↓ parasympathetic); rat: ↓ **CGRP, SP in DRG**; ↓ **PACAP innervation** → ↓ **efferent and afferent activity**.
- **CNS:** ↓ **5-HT and adrenergic spinal innervation** (sacral parasympathetic 5-HT preserved); **L1-L2 sympathetic preganglionic degeneration**, not L6-S1 parasympathetic. **Apomorphine (dopamine) frequency greater in aged rats** → aged CNS prone to **overactivity despite ↓ baseline function**. **UAB = unmet need** in ageing populations.
- **Detrusor:** **cholinergic/electrical responses unchanged** (though ↓ muscarinic density); **α-contraction ↑ (α1D ↑)** in most studies (one found none); **β-relaxation ↓** (↓ β density, ↓ cAMP) → **noradrenaline contracts the aged bladder but relaxes the young one**.
- **Antimuscarinics in the elderly:** **anticholinergic burden** → **cognitive decline**; a large case-control study linked **greater urological anticholinergic exposure (oxybutynin, tolterodine)** with **dementia risk** — **causality unclear** (symptoms may be **prodromal**).

KEY POINTS: AGING

- **Sensation and urethral pressure decline with age**; **reduced capacity reflects DO, not age** → ageing = **hypofunction**.
- **Detrusor responses to cholinergic/electrical stimulation unchanged with age**.
- **α-contraction ↑ and β-relaxation ↓ in old rats**.

Concept — **Ageing lowers sensation and drive but sensitises the circuits:** *nerves, afferents and β-relaxation decline while α-contraction and central excitability rise* — *reproducing DHIC (urgency with poor emptying) and explaining why anticholinergic burden is particularly hazardous in this group*.

Neuromodulation

Hypothesis of Mechanism of Action of Sacral Neuromodulation

- **Sacral, pudendal and posterior tibial neuromodulation** treat **refractory OAB and NOUR (pelvic floor dysfunction)** — the **apparent paradox**: one therapy both inhibits OAB and restores emptying.
- **Target = afferent axons in the roots** (benefit at intensities **below striated muscle activation**) → **somatic afferent inhibition of spinal sensory processing**: **retention resolves by inhibiting guarding reflexes**; **DO suppressed by inhibiting preganglionic neurons and interneuronal transmission in the afferent limb**.

Rationale for Neuromodulation to Facilitate Voiding

- Brain pathways must **switch off guarding reflexes** to void; **SCI** → **bladder-sphincter dyssynergia** (Fig. 103.43); subtler lesions (after **prostatitis or UTI**) may cause **idiopathic retention**.
- Before brain control matures (animals), **perineal pudendal afferents trigger efficient voiding** (bladder efferents on, outlet excitation off); **perineal touch inhibits the sympathetic guarding component**. **SNM may turn off outlet excitation**, and ↓ sphincter afferent input **disinhibits the bladder**.

Rationale for Neuromodulation to Inhibit the Overactive Bladder

- Inhibitory afferents from **sphincters, distal colon, rectum, anal canal, vagina, cervix, perineal skin** (Fig. 103.44: **SNS inhibits bladder-bladder and bladder-urethra reflexes**).

TWO AFFERENT INHIBITORY MECHANISMS

Feature	Interneuronal suppression	Direct preganglionic inhibition
Frequency	Most common	Less common
Site	Ascending limb → blocks bladder-to-PMC transfer	Bladder preganglionic neurons
Effect	Stops involuntary reflex voiding; spares voluntary voiding	Suppresses motor outflow directly — more effective
Triggered by	Somatic/visceral afferents	Pudendal electrical stimulation; anal canal/distal bowel — not penile/perineal touch

Pudendal Nerve Stimulation

- Pudendal nerve carries afferents of **all three roots S2–S4**; more peripheral → **avoids sciatic/sural activation** (thigh, calf, foot discomfort of S3 stimulation).
- **Course:** sacral plexus → on **piriformis** → **greater sciatic foramen** (below piriformis, with sciatic, inferior gluteal bundle, nerve to quadratus femoris) → **around ischial spine, superficial to sacrospinous ligament** → **lesser sciatic notch** → **ischioanal fossa** → **inferior rectal, perineal, dorsal nerve of penis/clitoris**.
- **Afferent pudendal stimulation inhibits micturition reflex, abolishes uninhibited contractions, ↑ capacity. Randomised crossover (tined leads): pudendal 63% vs sacral 46% symptom improvement; UUI reduction not significant (small numbers).**

Inhibitory and Excitatory Stimulation Frequencies of the Pudendal-Bladder Reflexes

- **Mechanism and programming parameters unstudied clinically. SCI cats: 3 Hz pudendal stimulation inhibits bladder; intermittent 20 Hz improves emptying. Continuous stimulation may fatigue the sphincter and accommodate the nerve; continuous vs intermittent outcomes unexplored.**

Activation of Somatic Afferents in the Foot

- **Transcutaneous footpad stimulation inhibits reflex bladder activity (cat) → FootStim: wearable home device, 3 h daily, ↓ UUI and urgency/frequency in refractory OAB (women).**

OnabotulinumtoxinA Neuromodulation

- **Blocks presynaptic ACh release (striated and smooth muscle). About 150 kDa single chain → cleaved to heavy chain (about 100 kDa) + light chain (about 50 kDa, zinc-bound) linked by a disulfide bond.**
- **Four steps: heavy chain binds nerve-terminal receptor → internalisation → light chain translocates to cytosol → blocks transmitter release. OBTX cleaves SNAP-25 (a SNARE, soluble N-ethylmaleimide-sensitive factor attachment protein receptor) → no vesicle fusion.**
- **Also afferent: ↓ urgency in NDO/IDO with ↓ TRPV1 and P2X3 in C fibres; ↓ SP, CGRP from sensory terminals; ↓ urothelial ATP; lipotoxin (liposomal OBTX) shows the urothelium is a target. Uses extended to pelvic floor spasticity, non-neurogenic OAB, BPS.**

LANDMARK COMPARATIVE TRIALS IN REFRACTORY IDIOPATHIC OAB WITH UUI

Trial	Comparison	Population	Result	Trade-off
ABC (Visco et al., 2012)	Intradetrusor OBTX vs oral antimuscarinic	Idiopathic OAB with UUI	Same reduction in daily UUI; OBTX more complete resolution	Antimuscarinic: more dry mouth; OBTX: more catheterisation and UTI

Trial	Comparison	Population	Result	Trade-off
ROSETTA (Amundsen et al., 2016, 2018)	OBTX 200 U vs SNM , 24 months	Women, refractory non-neurogenic OAB	Comparable efficacy; OBTX more often completely dry	OBTX: more UTI (note 200 U dose)

Concept — Neuromodulation works through afferents, not muscles: *stimulating somatic afferents inhibits spinal reflex processing, which suppresses the overactive bladder and releases the guarding reflex in retention — resolving the apparent paradox — while OBTX likewise acts on both efferent and afferent (and urothelial) signalling.*

FUTURE RESEARCH

- **Established newer therapies:** OBTX, β_3 agonists, SNM, PTNS. Next: "**omics**" (genomics, proteomics, transcriptomics, lipidomics, metabolomics) for **deeper, biology-based phenotyping** beyond symptoms.
- **Pharmacogenomics** → individualised therapy; **regenerative medicine and stem cells (e.g. mesenchymal)** for storage, emptying and urethral closure; **neuromodulation** of roots, tibial, foot and **saphenous** nerves, and **brain stimulation (TMS)** (already used in depression, schizophrenia, addiction, tremor).
- **Urinary microbiota** characterised in normals, OAB/UUI and IC/PBS; understanding needs **host responses** (urothelium, lamina propria, afferent/efferent pathways).
- **Questions:** prevention of LUTD; genotype-stratified ("personalised") pharmacotherapy; **biomarkers for phenotyping, prognosis and new targets;** susceptibility genes; **afferent modulation;** urothelium as target; **a treatment for underactive detrusor that makes self-catheterisation obsolete;** clinical value of regenerative medicine; host response to microbiota.